

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

PUBLICATION DATE **7.10.2020**

VERSION **v. 1.1**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2- Contact information of VPA Implementer (mandatory)

Appendix 3- Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11358
Title of Project	PoA GS11139- Access to Improved Cooking Solutions Programme- Efficient Kitchens in India-VPA01
Time of First Submission Date	16/12/2021 ¹
Date of Design Certification	07/04/2022
Version number of the VPA-DD	08
Completion date of version	29/06/2022
Coordinating/managing entity	ClimateCare Oxford Limited
VPA Implementer (s)	Greenway Grameen Infra Private Limited
Project Participants and any communities involved	N/A
Host Country (ies)	India
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A

¹ The LSC was first submitted under the PoA (GS11139) during the PoA design review on 19/03/2021

Project Cycle:

☒ Regular

☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission reduction	91,702	VERs
1. No Poverty	Estimated monetary savings computed from fuel saved.	21.51rupee/day ²	Rupee/household/day
5. Gender Equality	Average time saved by women for fuel collection	3.45	Hours/household/day
7. Access to clean energy	Number of people/households with access to basic service provided by the project device	41,729 (33,383 Jumbo, 2,086 smart and 6260 prime)	Number
8. Decent Work and economic growth	Total number of jobs split by gender	300 (200 male, 100 female)	Number of people

² Estimated based on survey finding on expenditure and using stove fuel saving percentage, the estimated amount of savings in then computed. This estimate vary from place to place and from time to time.

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

The purpose of the project is to manufacture, promote and distribute efficient fuelwood cooking stoves to domestic users across India where the VPA will be located. The proposed VPA will have the whole of India as it's the project boundary. The location of the VPA is within the proposed PoA boundary.

The project has been developed and is being implemented by Greenway Grameen Infra Private Limited, who manufactures the stoves. Currently there are three stove models being promoted by this project activity which includes the Prime, Smart stoves and Jumbo stoves. The stoves have different efficiencies as follows 28.9% for Prime stoves, 32.098% for Smart stoves and 38% for Jumbo stoves³. The stoves employ natural draft technology to achieve high thermal efficiency.

The project activity targets to distribute a maximum of 41,729 stoves including 33,383 Jumbo, 2,086 smart and 6,260 prime stoves which will displace baseline inefficient cooking devices that consume more non-renewable biomass. The stoves targets to benefit end-users who in the baseline were using inefficient biomass stove technologies. From the applied methodology (Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0), the baseline scenario has been determined as the consumption of non-renewable biomass using inefficient cooking devices to deliver thermal energy among household end-users.

The project activity has been designed with an incentive scheme that involves the use of a credit facility to enable end-users to purchase the stove. Such a facility is not available to the baseline technology users. This is an incentive mechanism which is being utilised to enable uptake of project stoves and also to discourage the use of the baseline stoves. The boundary covered by the VPA is the geographical boundary of India.

Goal and Objectives of the Project Activity

The goal of the project activity is to promote use of energy efficient fuelwood cookstoves which will significantly reduce use on non-renewable biomass for cooking leading to fuel savings. Use of the project technology will displace GHG emissions from the thermal

³ See Evidence 9- Stove Efficiency results folder

energy consumption of households and contribute towards reduced household air pollution.

Specific Objectives

1. To increase access to improved fuelwood cookstoves within the project area which will replace traditional inefficient biomass cookstoves used in the baseline.
2. To contribute towards improved health among end-users through the reduction of household air pollution.
3. To reduce greenhouse gas emissions associated with cooking with non-renewable biomass in the targeted populations.

The location of the project activity

The location of the project activity is the Republic of India. The Republic of India lies within Latitudes 8° 4' and 37° 6' north, and longitudes 68° 7' and 97° 25' east. The target project areas are the states in India.

Duration

The project start date is 29/10/2020 which marks the date when the first unit was sold under the project activity. The crediting period start date is 29/10/2020. The total length of crediting period is 15 years (5 years renewable twice).

Registration with other Carbon standards

The project activity is not registered with any other carbon standard, nor is it seeking registration under any other GHG programs, and hence there is no likelihood that double counting would occur. Additionally, each cookstove shall be assigned a unique serial number which gives a distinction on the end-users which will further ensure there is no double counting.

Carbon credits ownership

The legal ownership of the products generated belongs to ClimateCare Oxford Limited (CME). The end-users' consent to the transfer of carbon credits generated from using the project technology at the time of sale to the project implementor (Greenway Grameen Infra Private Limited), who in return has signed a contract with the CME to deliver all the VERs generated from the project activity.

Implementation plan

Greenway Grameen Infra Private Limited is the project implementor and will be directly responsible for production of the stoves, distribution and maintaining the sales database. They will also be responsible for collecting baseline information, co-ordinating the annual monitoring process and maintenance of the stoves.

The project activity's projection is to distribute 41,729 units (33,383 Jumbo, 2,086 smart and 6260 prime stoves) over a 5-year period, which will be installed within end-user's premises located within the target area. The expected annual average emission reductions are 91,702 tCO₂e, and the total emission reductions during the first crediting period are 458,508 tCO₂e.

Host Country regulations on emission reductions

The host country has not set any regulations setting a cap on the emission reduction generated by projects. There is also no possibility by the host country to trade emissions that include the scope of the proposed project.

A.1.1. Eligibility of the project under approved PoA

Table 2: VPA eligibility under the PoA

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1.	<i>Geographical boundaries of V/CPAs must be consistent with the geographical boundary of the PoA;</i>	The VPA shall be located within the PoA boundary of India	The VPA is located in the Republic of India which is within the PoA boundary.
2.	<i>Conditions to avoid double counting of Impacts, such as unique identifications of product and end user locations (e.g. programme logo);</i>	Identification of each device through unique identification number. Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another PoA.	Greenway stoves have a unique serial number that ensures they are uniquely identifiable to this project. Sample pictures of stove with serial numbers have been provided to the VVB (<i>Evidence 4-Sample pictures of stoves serial numbers</i>). Further, each end user shall have only one registered project stove to ensure there is no double counting. An updated sales database will be maintained and will record the number of stoves sold to each end-user. Only one stove shall be eligible for crediting. The PP has also signed a funding agreement with the CME which outlines that the VPA is only registered to the PoA (<i>Evidence 2</i>).
3.	<i>Conditions to confirm that V/CPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered;</i>	Confirmation that the VPA is neither registered with other carbon standards	The CME has declared that the VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA. The PP conducted a search on the Gold Standard, CDM and VERRA registries to check the following: (i) Registration of the project with other standards (ii) similar projects registered

			within the same project boundary (iii) projects using the same project technology as the project and (iv) whether the project activity had been deregistered by any of the standards. on the search was conducted on 03/09/2021 and the PP can confirm that the VPA is not registered with other Carbon Standards. However, from the search it was established that there are other programs that were found to be making use of the Greenway stoves as follows: Gold Standard projects: GS10818: 1. Dissemination of Improved Cookstoves in India by Greenway - PoA 2. GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway - VPA001 3. GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway - VPA002 4. GS11242- Fair Climate Programme for Sustainable Household Energy (PoA) CDM 1. CDM 9181: MicroEnergy Credits – Microfinance for Clean Energy Product Lines –
--	--	--	---

			India (unfccc.int)- POA
			2. MicroEnergy Credits POA – Grameen Koota Indoor Air Pollution Campaign
			3. MicroEnergy Credits PoA – CPA 02
			4. MicroEnergy Credits PoA – CPA 03
			5. MicroEnergy Credits PoA – CPA 04
			6. MicroEnergy Credits PoA – CPA 06
			7. MicroEnergy Credits PoA – CPA 07
			8. MicroEnergy Credits PoA – CPA 08
			9. MicroEnergy Credits PoA – CPA-19- Clear Sky Partners
			10. MicroEnergy Credits PoA – CPA 11
			11. MicroEnergy Credits PoA – CPA 12
			12. MicroEnergy Credits PoA – CPA 13
			13. MicroEnergy Credits PoA – CPA 15
			14. MicroEnergy Credits PoA – CPA 16
			15. MicroEnergy Credits PoA – CPA 17
			16. MicroEnergy Credits PoA – CPA 18
			17. MicroEnergy Credits PoA – CPA-21- Clear Sky Partners
			18. MicroEnergy Credits PoA – CPA-24- Clear Sky Partners

		19. MicroEnergy Credits PoA – CPA-26- Clear Sky Partners 20. MicroEnergy Credits PoA – CPA-29- Clear Sky Partners 21. MicroEnergy Credits PoA – CPA 31 22. MicroEnergy Credits PoA – CPA 35 23. MicroEnergy Credits PoA – CPA 36 24. MicroEnergy Credits PoA – CPA 37 To ensure there is no double counting, Greenway have a robust stove identification system where each stove in each project is clearly distinguished by a unique serial number and cannot be included in any other project. The projects are implemented by different project developers and each project maintains a separate and distinct database hence avoiding any risk of double counting. The CME can also confirm that the VPA is not a deregistered VPA which is being introduced again. Although there are other projects whose boundaries overlap with the proposed project, the CME confirms that appropriate procedures have been put in place to distinguish each project database and separate each unique stove serial number being included in their
--	--	--

			respective project databases and this action prevents double counting other mitigation outcomes (See ICS 6 for more details).
4.	<i>Specification of the technology/measure such as the level and type of service, as well as performance specification based on, inter alia, testing/certification;</i>	The VPA shall involve the distribution and sale of highly efficient biomass burning stoves with 20% efficiency and above. Each VPA will involve the distribution and installation of efficient cookstoves to households and/or communities currently cooking with non-renewable biomass.	As per the lab test results provided by the PP, the efficiency of the 3 Greenway stoves was tested and has been confirmed as follows: Smart Stove (32.098%), Jumbo Stove(38%) and Prime stoves (29.8%). Copy of test results provided to VVB (<i>Evidence 9</i>). The project sells stoves to households only.
5.	<i>Conditions to check the start dates of V/CPAs through documentary evidence;</i>	The start date of the VPA shall not be before the PoA stakeholders' consultation date.	The PoA design consultation was initiated on 14/10/2020, and it's also the date when the first stakeholder consultation meeting for the VPA was held. The VPA start date is 29/10/2020 which marks the date when the first distribution of the project stoves started. The stove database shows the actual distribution dates for the stoves and PP has also submitted supporting sample agreements signed between end-users and Greenway Grameen Infra Pvt. Ltd. As proof of start date (see Greenway-Start date and warranty evidence) and the provided stakeholder's consultation report which shows the date when the consultations were done.
6.	<i>Conditions to ensure compliance with the applicability of the applied methodologies, the applied</i>	The VPA shall meet the applicability requirements of the methodology "Technologies and	The VPA applies an approved Gold Standard methodology "Technologies and Practices to Displace

	<i>standardized baselines and the other applied methodological regulatory documents</i>	Practices to Displace Decentralized Thermal Energy Consumption Version 4.0" and GS4GG Principles and Requirements.	<i>Decentralized Thermal Energy Consumption Version 4.0"</i> . The applied methodology is applicable to the VPA since the methodology requires that only activities that introduce technologies and/or practices that and non-domestic premises can apply it. Since the project introduces stoves technologies that reduce greenhouse gas (GHG) emissions from the thermal energy consumption of households, the VPA therefore meets this condition.
7.	<i>Conditions to ensure that V/CPAs meet the requirements for demonstration of additionality</i>	Each VPA within the PoA shall demonstrate additionality through the Gold Standard additionality criteria or the latest version of the CDM Methodological Tool for the demonstration and assessment of additionality.	Assessment of additionality has been done at VPA level as outlined in section B.5 of the VPA-DD. Documentation: VPA-DD
8.	<i>Conditions to ensure no diversion of official development assistance;</i>	Confirmation of non-involvement of public funding or ODA from Annex I Parties in VPA	The Project activity have signed and ODA declaration form. The signed form has been made available to VVB and uploaded on SustainCert app.
9.	<i>Target group (e.g. domestic/commercial/industrial, rural/urban, grid connected/offgrid), and where applicable, distribution mechanisms (e.g. direct installation)</i>	Target groups should be identified	The VPA distributes improved fuelwood cooking devices to domestic end users and the distribution mechanism is outlined in section A.1 of the VPA-DD.
10.	<i>Conditions related to sampling requirements for the PoA</i>	Sampling requirements be outlined in section B.2 of the PoA shall be applied for the VPAs included into the programme.	The VPA will apply the latest Guidelines for sampling and surveys for CDM project activities and programmes of activities and the sampling process has been outlined in section B.7.2 of the VPA-DD.

11.	<i>Conditions to confirm that technologies in V/CPAs are eligible</i>	All VPAs to meet eligibility requirements specified in section B.3 of the PoA-DD whereby the stoves utilise non-renewable biomass as feedstock and will use fuelwood as the fuel type	The VPA is installing three models of efficient fuelwood cook stoves whose efficiency is as follows: Smart Stove (32.098%), Jumbo Stove (38%) and Prime stoves (29.8%) which is more than the minimum efficiency of 20% required under the methodology. The details of the stoves have been explained in section A.3 of this VPA-DD. The stove specifications have been provided to the VVB. The fuel being used is non-renewable biomass feedstock and fuelwood as the only fuel type.
12.	<i>Conditions to be met by each VPA regarding SDG outcomes assessment</i>	Each VPA shall conduct an SDG Outcomes assessment and shall positively contribute to the sustainable development goals identified in the PoA-DD.	An SDG outcomes assessment has been done in the VPA-DD in section B.6
13.	<i>Conditions to be met by each VPA regarding safeguarding principles</i>	Each VPA shall conduct a safeguarding principles assessment in line with the Gold Standard Safeguarding principles assessment requirements.	A safeguarding principles assessment has been done in the VPA-DD as detailed in Appendix 1
14.	<i>Legal Ownership</i>	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated.	End users shall sign a Carbon Waiver Form during purchase which transfers the rights of the carbon credits to Greenway Grameen Infra Private Limited.
15.	<i>On-going financial need</i>	<i>Each VPA to demonstrate on-going financial need</i>	Carbon revenues is utilised by the project through a revolving fund and stove subsidy which has been set up to help users to purchase the stove affordably and pay for the stove through a monthly repayment plan. This method helps end users afford

			the stove. Without the funds, end users will be able to afford the stove. See section B.5.2 below for more details.
--	--	--	---

Eligibility with GS4GG Requirements

The project activity falls under section 3.1.1 of *GS4GG Principles & Requirements* (Version 1.2), section 3 of *GS4GG Community Services Activity Requirements* (Version 1.2) and section 5 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements* (Version 2.0) with the following eligibility criteria:

Table 3: Demonstration of eligibility with GS4GG requirements

Eligibility Criteria	Justification
(a) Types of Projects Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Section 3.1.1 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Pre-identified CSA project types are a) Renewable energy; b) End-use energy efficiency; c) Waste management and handling; d) Water, sanitation, and hygiene (WASH). Section 5.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0)	The project activity involves the distribution of improved cookstoves among household end-users. Therefore, the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end." The project activity falls under section 5.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical

The Following Project types are eligible for issuance of Gold Standard VERs: a) Renewable Energy Supply; b) End-Use Energy Efficiency Improvement; c) Waste Handling & Disposal; d) Land Use and Forests. As per section 4.1.3 of <i>GS4GG Principles & Requirements</i> (Version 1.2), "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it, or it's referenced in the Gold Standard Product Requirements".	intervention is required at the user end." of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0). The project type is "End-Use Energy Efficiency Improvement" and the Impact Quantification Methodology to be applied is "Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v4.0". Method 1 given in the methodology for calculating emission reduction for improved biomass cookstoves is applied.
(b) Location of Project: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects may be located in any part of the world. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states	The project activity shall be located in the Republic of India (refer to section A.2 of the VPA-DD for detailed project area and boundary). There are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in India, thus based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country.

have mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	
(c) Project Area, Project Boundary and Scale: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. To avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall	The project activity location is the Republic of India (refer to section A.2 of the VPA-DD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in India, thus based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. Greenway Grameen Infra Private Limited is implementing another Gold Standard Project which uses their project stoves. In addition, the Greenway stoves are also used in another CDM registered project. However, each project maintains a separate and distinct database for each project. Each stove in each project is clearly distinguished by a unique serial number and cannot be included in any

demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects). Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 9.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) GSVER Projects may be registered as 'large scale', 'small scale' (for the applicability of methodologies and tools only) or 'microscale'. Scale is defined in the relevant Activity Requirements or where these do not exist then per following paragraphs. Section 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) All Projects exceeding the small-scale thresholds are defined as large scale. Small scale projects are defined in accordance with CDM project standard for project activities, as below;	other project (see section B.2 and parameter ICS 6 for detailed explanation). The project activity boundary is defined based on the methodology "<i>Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC)</i> (Version 4.0)" and is limited to the Republic of India (refer to section B.3 of the VPA-DD). The project activity is defined as large scale since the expected annual energy savings is 298.13GWh which exceeds 180 GWh.
---	---

(a) Type 1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th). (b) Type 2: End-use energy efficiency project improvement: activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period. In this context, for project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh(th) per year saving; (c) Type 3: Other project activities: project involves technologies such as Safe Water Supply, Waste management, etc. not included in Type I or Type II that result in GHG emission reductions not exceeding 60,000-ton CO₂e per year in any year of the crediting period.	
(d) Host Country Requirements Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The project activity location is the Republic of India (refer to section A.2 of the VPA-DD for detailed project area and boundary). The host country has not set any mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.). Based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) India is an eligible host country. A detailed assessment on the project's compliance with the applicable host Country's legal,

	environmental, ecological, and social regulations is summarized in tables 17.
(e) Contact Details Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	The Project Participants' contact details are provided in Appendix 2 of the VPA-DD. The project implementor is Greenway Grameen Infra Private Limited and is legally registered to operate in the host country as proof that it is in good standing. The company's certificate of incorporation is provided as supporting evidence (<i>see Greenway and ClimateCare Oxford Limited Certificates of incorporation</i>).
(f) Legal Ownership Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested	All end users at the time of purchase of the Greenway stove, sign a carbon rights waiver which transfers the carbon rights to Greenway Grameen Infra Private Limited. By signing the waiver, the rights to the carbon are transferred to Greenway Grameen Infra Private Limited and they remain uncontested. Subsequently, Greenway Grameen Infra Private Limited has signed an Emission Reduction Purchase Agreement where they will transfer the carbon rights eventually to ClimateCare Oxford Limited

legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising. Section 3.1.4 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	(CME). Refer to section A.1.2 of the VPA-DD for further details. Moreover, during the local stakeholder consultation meeting, the project implementer informed the stakeholders present the need for ownership of the carbon rights, and it was determined that the project o had the undisputed carbon legal ownership.
(g) Other Rights Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known	The project activity involves distribution of improved biomass cookstoves. The project activity does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.

disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	
h) Official Development Assistance (ODA) Declaration Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification. Section 6.1.1 and 6.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Projects are ineligible for carbon crediting under Gold Standard if the ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly,	The host country (India) is named in the OECD ODA recipient list⁴. The project developer has signed the ODA declaration and confirms that no ODA has been provided for the project activity (see provided <i>_GS11139 and GS 11358 ODA Declaration documents</i>)

⁴ [DAC List of ODA Recipients for reporting 2021 flows June 2021.xlsx \(oecd.org\)](#)

to the donor country providing ODA support. Project Developer submitting a Project located in a country named by the OECD Development Assistance Committee's ODA recipient list shall sign and submit the ODA Declaration.	
i) Eligible Greenhouse Gases Section 4.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version2.0)</i> Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	The project activity only considers the reduction of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O) emissions for Gold Standard crediting. Refer to section B.3 of the VPA-DD for further details.

Corrective Action Requests (CARs) and Forward Action Requests (FARs)

2 Forward Action Requests (FARs) and 1 Corrective Action Request were raised by the Gold Standard during the Preliminary review, and these have been addressed as discussed below:

Likely CAR # 1: The start date shall be justified in line with the definitions of the start date. For distributed technology projects only, the start date is defined as the implementation of the first unit under the project. The relevant evidence for the start date shall be provided at the time of validation. The VVB shall provide opinion on the same at the time of validation.

The PP can confirm that the project start date is 29/10/2020 which marks the date when the first unit under the project was sold. PP has submitted sample agreements signed between the project implementer and end-users upon sale of the units as evidence of the project start date (see provided start date and warranty evidence document).

FAR # 1: The PD shall supply supporting data for all parameters in time for validation/design review, or allocation may be delayed. This includes and is not limited to: ER spreadsheets, individual study calculations, survey results, study reports etc.

The PP can confirm that a baseline scenario survey was conducted and the baseline data and supporting evidence has been provided. A Kitchen performance test is being planned and the findings will be presented prior to the first verification (See cookstove baseline survey responses spreadsheet). The ex-ante emission reduction calculations have also been done in line with the methodology as provided under the Greenway ER calculation spreadsheet. Other supporting documents for the parameters have also been submitted for GS review.

FAR # 2: The new version (version 4.0) of the applied methodology (TPDDTEC) is now published on the GS website. The PD shall note that there is no grace period for the last version (version 03) and the new project shall use the latest version 04 of the applied methodology.

As the preliminary review already started with the older version, hence the following requirement applies. "GHG EMISSIONS REDUCTION & SEQUESTRATION PRODUCT REQUIREMENTS § 8.2.2 | The methodology and tool version applied at the time of first submission (preliminary review) may be used for Design Certification as long as the Project is submitted for validation within six months after the time of first submission for

Preliminary Review. If this condition is not met, the latest version of the methodology and applicable tool(s) available at the time of submission for validation shall be applied". Hence the above requirements shall be taken care of during the validation.

The PP can confirm that the project has upgraded to the new methodology TPDDTEC (Version 4.0) as required. This has been included in the VPA-DD and the parameters set out in the methodology are applied in the Ex-ante calculations. The project activity is eligible under the latest version of the methodology and the applicable emission reduction calculation method is method 1.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

All end users at the time of purchase of the Greenway stoves, sign a carbon rights waiver which transfers the carbon rights to Greenway Grameen Infra Private Limited. By signing the waiver, the rights to the carbon are transferred to Greenway Grameen Infra Private Limited and they remain uncontested. Subsequently, Greenway Grameen Infra Private Limited has signed an Emission Reduction Purchase Agreement where they transfer the carbon rights eventually to ClimateCare Oxford Limited (CME).

A.2. Location of project

The project boundary for the proposed VPA is the Republic of India. The project activity will be implemented in all the states in India. The geographical coordinates for India are Latitudes 8° 4' and 37° 6' north, and longitudes 68° 7' and 97° 25' east ⁵ as shown in map below.

⁵ <https://www.india.gov.in/india-glance/profile>



Figure 1: The geographical map for the Republic of India

Source: [Wikimedia](https://commons.wikimedia.org/wiki/File:India_map.svg)

The Greenway Appliances factory is located at 501-502 Makani Centre, 35th Road, TPS III, Bandra West, Mumbai 400050.

A.3. Technologies and/or measures

The VPA involves the dissemination of improved fuelwood cookstoves within the Republic of India. The improved cookstoves are manufactured by Greenway Grameen Infra private and they have been designed to improve the combustion efficiency, thereby reducing the fuelwood consumption and emissions generated while cooking. As the combustion of fuel in these stoves occurs at nearly stoichiometric conditions, the amount of pollutants

emitted are also reduced. The technical specifications of the three stove models (Smart, Jumbo and Prime) disseminated under the project are given below:

Parameters	Smart Stove	Jumbo Stove	Prime stoves
			
Manufacturer name	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt
Size:	9.8" x 7.6" x 11.6"	12.4" x 10.6" x 11.6"	10" x 9" x 11.9"
Materials:	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles
Loading Capacity:	25 kg	40 kg	40kg
Secondary Air Induction Mechanism:	Yes	Yes	Yes
Warranty:	1 year	2 years	2 years
Fuel Type	Ergonomic front loading design	Ergonomic front loading design	Ergonomic front loading design
Operational Lifetime	72 months	72 months	72 months
Fuel type	Fuelwood	Fuelwood	Fuelwood
Application/service level	Domestic cooking	Domestic cooking	Domestic cooking
Energy output	1.34 KW	2.32 KW	TBC ⁶
Efficiency	32.098%	38%	29.8%

These stoves models, which are essentially the same design but of different sizes, have been developed by Greenway Grameen Infra Private Limited. To determine their efficiencies, the stoves were tested as follows:

- Prime and Smart stoves by Indian Institute of Technology
- Jumbo stove by the Kenya Industrial Research and Development Institute (KIRDI⁷)⁸.

In the future, Greenway Grameen Infra Private Limited might develop additional stove models which have higher performance and once developed, they can be included in the VPA.

⁶ To be determined before 1st verification

⁷ [Home - KIRDI](#)

⁸ See Evidence 9- Stove Efficiency results folder

Contribution to SDG benefits

The Greenway stoves are durable and efficient innovative stove technologies that are set to deliver fuel savings of about 56.8%. The beneficiary households will realize multiple benefits from the use of this project technology in the following ways:

- **Improved health:** Using traditional inefficient biomass cookstoves create negative health impacts associated with indoor air pollution along with eye & skin irritation and other ailments. Through the introduction and use of the improved project cookstoves, the households will benefit from improved health benefits due to reduced IAP related illnesses (SDG 3)
- **Environment & Climate Change:** Traditional inefficient biomass cookstoves consume more fuel, typically non-renewable firewood or charcoal. Emissions associated with these stoves are a significant contributor to climate change. The project activity promotes use of improved cookstoves which will reduce the demand for non-renewable biomass. This will consequently reduce greenhouse gas emissions associated with use of biomass fuels for household cooking (SDG 13).
- **Gender and Opportunity:** Cooking on traditional biomass inefficient stoves consumes a lot of fuel leading to constant need for sourcing of fuel. Fuel collection requires more time and effort, especially by women and children which takes away their productive hours leading to time poverty and lost opportunity. Through reduced fuel usage as a result of using the project technology, the project activity will contribute to a reduction in time required to fetch fuel, which can be spent on other income generating activities and study for women and children (SDG 4 and 5).
- **Monetary gain and Time savings:** Families spend a significant portion of their time and income on fuel collection and purchase. As fuel becomes scarce, expenditure on fuel becomes more significant due to the increase in fuel prices and this leads to more household income being spent on meeting the household energy needs. This leads to a significant reduction in household disposable income. By reducing the fuel usage through the project technology, the project activity will contribute to a reduction in the fuel cost for the households which can be spent on other household needs (SDG1).
- **Job creation:** The project activity leads to job creation in the stove production and supply chain. This is expected to benefit many people especially locals within the host country in the project areas (SDG 8).

A.4. Scale of the project

The project is a largescale project activity. As per the product activity requirements, end-use energy efficiency improvement projects are within the largescale limit if they have an annual energy saving of more than 180 GWh. As per the energy savings calculations, project activity will have an annual energy saving of 298.13 GWh which is more than 180 GWh (see *GS11358-Greenway ER calculation tab Energy calculation*).

A.5. Funding sources of project

There is no public funding utilized by the project. To affirm the same, the project developer has signed the ODA declaration form which has been submitted to the Gold Standard as a supporting document.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The applied methodology by the VPA is "Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC), Version 4.0"

The VPA has also applied the following latest versions of the tools and guidelines:

- Tool 30: Calculation of the fraction of non-renewable biomass, version 3.0"
- GHG Emissions Reduction & Sequestration Product Requirements, Version 2.0
- Community Services Activity Requirements, Version 1.2
- Programme of Activity Requirements, Version 1.2
- Stakeholder Consultation and Engagement Requirements, Version 1.2
- Stakeholder Consultation and Engagement Guidelines, Version 1.2
- GS4GG Principles & Requirements, Version 1.2
- Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0
- Usage Survey Guidelines, version 2.0.

These are the latest versions of the methodology and tools which have been applied at the time of first submission of the project to Gold Standard.

B.2. Applicability of methodology (ies)

The project activity applies the GS methodology “Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0”, since the methodology is applicable to technologies and or practices that reduce or displace greenhouse emissions from the thermal energy consumption of households and non-domestic premises, it applies to this project which distributes domestic non-renewable biomass cookstoves.

The VPA involves the dissemination of improved cookstoves across India. The cooking devices being distributed are of high combustion efficiency and help in saving fuel while providing the same level of cooking service to households.

The VPA meets the applicability criteria of the methodology as follows:

Eligibility of the project under the applicable methodology (TPDDTEC v.4.0)

Applicability condition	How Applicability condition is met by VPA
Section 2.1.1. provides that this methodology is applicable to project activities that introduce technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and/or residential, institutional, industrial, or commercial facilities. Throughout the methodology the term ‘technology’ should be read as the single or multiple technologies and/or practices applied in the project activity.	The project activity entails distribution of efficient biomass cookstoves. Use of these technologies will displace GHG emissions from the thermal energy consumption of households. The project is therefore eligible and emission reductions shall be calculated using method 1.
Section 2.1.2: Where there is no installation of improved devices and project claims emission reductions from improved practices only, project shall provide a detailed discussion of the chosen monitoring approach to demonstrate that quantified emission reductions result exclusively from the practices introduced by the project activity.	The project activity entails distribution of improved biomass cookstoves with higher efficiency compared to the baseline stove technologies and emissions reductions shall be claimed from improved efficiency.

Section 2.1.3: Project may involve progressive distribution of technology where implementation of the technology may occur in a gradual manner and adoption can increase over the project's crediting period.	The project activity is projected to progressively distribute 41,729 stoves among end-users within the project boundary over the 5-year period. The implementation of the project requires sound logistical and finance planning hence the technologies will be distributed progressively until the target is reached.
Section 2.2.1 (a) Project shall choose a technology design that has predictable performance in that it is proven to be efficient and durable under field conditions; for cookstoves, the rated thermal efficiency shall be at least 20%	The efficiency of the 3 stoves is as follows: Smart Stove (32.098%), Jumbo Stove (38%) and Prime stoves (29.8%), which is above the minimum efficiency of 20% required under the meth. In line with para 3.7.3 and 3.7.4 of the methodology TPDDTEC, version 4, since the difference in the efficiency of the stoves is more than 5%, the project technologies has been treated as independent project scenarios in ex-ante estimations and the same will be applied during the monitoring.
Section 2.2.1 (b) The technology shall have continuous useful energy output of less than 150kW per unit	The annual energy saving by the project stoves is less than 150kW as shown in table on section A.3.
Section 2.2.1(c) The project activity is implemented by a project developer and can include additional project participants listed in Appendix 2 of the PDD template. The individual households and institutions may be represented collectively by community organizations, etc., but do not individually act as project participants.	The project activity is implemented by Greenway Grameen Infra Private Limited who is the project developer. Individual households are not project representatives in the project.
Section 2.2.1(d) The project developer must design incentive mechanism(s), which should be effective as fast as possible, for the elimination of inefficient baseline stoves that are replaced by the	The project activity will be distributing the project stoves using a credit facility to enable users to purchase the stove. This facility is not available for the purchase of the baseline technologies (See Section A.1

project cooking devices and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation	of the VPA-DD). This serves as an incentive mechanism that is utilised to enable uptake of project stoves and to discourage the use of the baseline stoves. The VPA will also monitor the extent to which the baseline technology continues to be used and where found, an appropriate adjustment carried out to account for the use of the baseline technology.
Section 2.2.1 (e) To avoid double counting or double claiming, the project developer must: i. clearly communicate its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties by contract or clear written assertions in the transaction paperwork: all other project participants; project technology manufacturers; and retailers of the project technology other renewable fuel in use; and ii. inform and notify the end users that they cannot claim emission reductions from the project, and exclude from the project activity, cooking devices included in any other voluntary market or CDM project activity/PoA and strive not to displace the cooking devices of another CDM or voluntary project/PoA. See data and parameters not monitored, Avoidance of double counting or double claiming with other mitigation actions, for details on this demonstration	i. The VPA communicates to the end users about carbon rights during purchase of the stove and the end users sign a carbon rights waiver to transfer the carbon rights to the VPA developer (Greenway Grameen Infra Private Limited). Subsequently, Greenway Grameen Infra Private Limited has signed an Emission Reduction Purchase Agreement where they transfer the carbon rights eventually to ClimateCare Oxford Limited (CME). ii. End-users are informed about how carbon credits are utilised to support the stove project, including the credit facility. They are also informed that they cannot, therefore, claim emission reductions from the project. The project stoves are identified by their serial numbers through which the stoves are uniquely traceable to the end users based on the sales database records. The CME ensures that the sales database is well maintained and current and only the serialised stoves in the sales database are included for emission reduction calculations. This way, double counting is avoided for the project stoves.

Section 2.2.1 (f) Project activities making use of solid fossil fuel in the project scenario or other improved fossil fuel cookstoves meeting certain conditions described in the footnote to Table 1 (e.g. switch from three-stone fire biomass stoves to LPG stoves) may only claim emission reductions for energy efficiency improvement aspect and shall assume the same baseline and project fuel for emission reduction calculations	Not applicable to the project. The project only uses non-renewable biomass.
Section 2.2.1 (g) Project activities making use of a new solid biomass feedstock in the project situation (e.g. switch to green charcoal or renewable biomass briquettes) must comply with relevant specific requirements for biomass related project activities, as defined in the latest version of the Community Services Activity Requirements. The specific requirements apply to both plantations established for the project activity and/or existing plantations that will supply biomass feedstock	Not applicable to the project. The project only uses non-renewable biomass.
Section 2.2.1 (h) Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases emitted by the project fuel/stove combination are estimated with adequate precision. Furthermore, for projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (for example, changing from a stove with chimney to improved stove with no chimney), indoor air pollution (IAP)	The Greenway stoves are portable, it is possible for the cooking to be moved from outdoor to indoor. The project stoves have been tested for PM2.5 as a way of determining their indoor air pollution levels. This will be compared with the PM 2.5 levels of the baseline stoves to be determined from published literature. Furthermore, the project will not reduce ventilation in the kitchens as they are improved stoves that will actually

levels shall not worsen in the project compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions. This may be demonstrated before project Design Certification or during project operation using the certification resulting from of a manufacturer's test, report of field testing of the technology's PM 2.5 and carbon monoxide (CO) emissions, report of lab testing of the technology, or results of modelling of the technology's operation under field conditions. If none of these are available, reference from published literature or report by independent agencies may be used as evidence, provided it is not more than 5 years old.	contribute to improved indoor air conditions among beneficiary households.
Section 2.3.1 The project shall not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply or fuel supply or use. The project shall document the national, regional and local regulatory framework for provision of thermal energy services of the type the project provides in the project boundary (parameter ICS 7).	There are no regulations in India that require households to adopt efficient fuelwood stoves. However, The Ministry of New and Renewable Energy approves models of portable improved biomass cookstoves, which users can purchase (See Section B-4 of the VPA-DD). One of the approved stoves models is the Greenway stove (see evidence; List of Approved Cookstoves by the Ministry).
Section 2.3.2 If the expected technical life of project technology (parameter ICS 3) is shorter than the crediting period, the project developer shall describe measures to ensure that end users are provided replacement technology of comparable quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee. If neither of	The technical life of the project stove is 6 years which is more than the GS crediting period. Therefore, there will be no replacement of the technology before the end of the crediting period.

the prior conditions can be demonstrated, no emission reductions can be claimed for the technology after its technical life has ended.	
--	--

B.3. Project boundary

The project boundary is specified in Section 3.2 of the applied GS methodology as follows:

- a. The project boundary is the physical, geographical sites of the project technologies/practices including the fuel collection and production area.
 - i. Where the baseline fuel is woody biomass (including charcoal), the project boundary also includes the area within which this woody biomass is grown and collected.
 - ii. For projects using processed fuels, this boundary also includes the baseline and project fuel production (e.g., charcoal, plant oil) and solid waste and effluents disposal or treatment facilities associated with fuel processing.
 - iii. In cases where the project activity introduces the use of a new biomass feedstock into the project situation, the fuel production and collection area are the area within which this new biomass is produced, collected and supplied.
- b. The target area is the region(s) or town(s) where the considered baseline scenario(s) are deemed to be uniform across political borders. This area could be within a single country, or across multiple adjacent countries. The target area provides an outer limit to the project boundary in which the project has a target population.

The aim of the project activity is to introduce use of improved firewood cookstoves which will promote displacement of inefficient firewood cookstove technologies used in the baseline for cooking.

The boundary for the project activity is therefore defined as the end-user households where the ICS will be installed and the environs within which the project technologies are utilized.

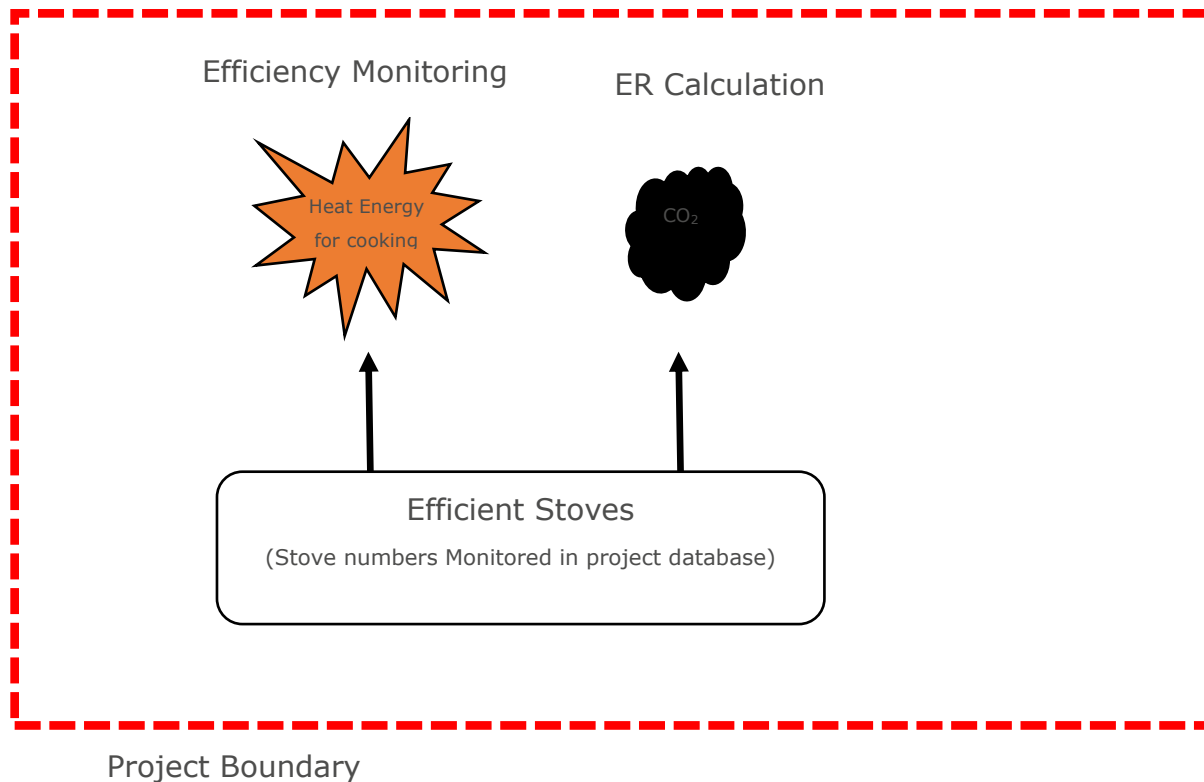
The target area for this project activity is defined as the areas in which the project activity is to be implemented within the Republic of India, identified as the project boundary. The target area where the baseline scenario is deemed to be uniform across India.

The summary of the baseline fuels accounted for during the baseline survey conducted with 263 households is as follows:

Primary fuel	Outcome	%
Crop residue	1	0.4%
Dung cake	7	2.7%
Wood	244	92.8%
Kerosene	1	0.4%
LPG	10	3.8%
Total	263	100%

From the findings it is evident that the main baseline fuel is biomass fuels and the greenhouse gases included in or excluded from the project boundary based on the baseline fuels are shown in table below.

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Delivery of thermal energy (fuelwood, biomass residues and animal dung, kerosene and LPG for cooking, majorly on inefficient cookstoves)	CO ₂	Yes	Important source of emissions
		CH ₄	No	Minor emission source excluded as a conservative measure
		N ₂ O	No	Minor emission source excluded as a conservative measure
Project scenario	Delivery of thermal energy (fuelwood using efficient cookstoves for cooking)	CO ₂	Yes	Important source of emissions
		CH ₄	No	Minor emission source excluded as a conservative measure
		N ₂ O	No	Minor emission source excluded as a conservative measure



B.4. Establishment and description of baseline scenario

According to the methodology applied by the project activity, TPDDTEC, version 4.0, the baseline scenario is defined by the existing baseline technology/practice use and fuel consumption patterns for the type of service provided by the project technology in the population targeted for adopting the new project technology. Hence, this “target population” is a representative baseline for the project activity.

The selection and description of the baseline scenario has been informed by the findings of the Baseline Scenario Survey conducted for the project activity. The baseline scenario survey was conducted with 263 households who were randomly selected from the project target areas, and who were not using the project technology. The 263 surveys were conducted between 07/09/2020 to 13/12/2020 by the project implementor (Greenway).

The surveys were done in 10 states in India which were drawn from the six regions of India. The 10 states selected were areas where the project had started distributing the stoves or where the project envisaged to distribute stoves in near future. The 10 states were found to be representative of the project boundary since it was not possible to conduct surveys in all the states in India due to the geographical spread and the

Covid 19 pandemic which was limiting movements and social interactions in the host country. The Google forms online platform was used to collect the baseline data (see *provided Greenway-Google form survey template*). The summary of the baseline surveys by state is as summarised in table below.

Table 4: Summary of baseline surveys by State

State	No of surveys
Assam	10
Bihar	24
Chhattisgarh	32
Gujarat	50
Jharkhand	2
Karnataka	125
Madhya Pradesh	1
Maharashtra	8
Tamil Nadu	6
Uttar Pradesh	5

Source: Baseline survey data spreadsheet tab 2020-baseline survey

Out of the 263 surveys conducted, 244 households (92.8%) were using firewood for cooking as the primary fuel, other fuels used included crop residues, dung cake, kerosene and LPG as summarised below.

Table 5: Baseline fuels used by end-users

Primary fuel	Outcome	%
Crop residue	1	0.4%
Dung cake	7	2.7%
Wood	244	92.8%
Kerosene	1	0.4%
LPG	10	3.8%
Total	263	100%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

The survey also assessed the secondary fuels used by the households. 27 households reported using secondary fuels and from the findings, the predominant secondary fuel was firewood used by 16 households (59.3%). The summary of the secondary fuels is as shown in table below.

Table 6: Secondary fuels used in the baseline

Secondary fuels	Outcome	%
Charcoal	1	3.7%
crop residue	8	29.6%
dung cake	2	7.4%
wood	16	59.3%
Total	27	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

From the baseline survey conducted by the project, it was determined that the targeted communities use mostly non-renewable biomass in traditional inefficient stoves or conventional devices without a grate or a chimney. The default efficiency of such stoves is 10%⁹. However, since other stove technologies were identified during the baseline surveys and their efficiency varies as summarised in table 7 below, a weighted average efficiency value of 15.5% has been calculated and is applied in emission reduction estimation. The project stoves are only for cooking and no cross effects are expected as a result of the project.

The baseline scenario has therefore been determined as, the consumption of non-renewable firewood using traditional inefficient stoves or a conventional device without a grate or a chimney to meet thermal energy requirements for household cooking. From the baseline survey findings, 174/263 households (66.2%) were using traditional 3 stone firewood stove as the primary baseline stove. The summary of the stoves used is as summarized on table below.

⁹ https://cdm.unfccc.int/filestorage/3/B/X/3BX1V2NC87TAWL0EKYHJQ5DRZ6UF94/EB108_repan10_AMS-II.G_clean.pdf?t=ZGt8cjdjbGtifDCvfLfSPsyQCDVGg91WG4 , table 9.

Table 7: Primary baseline stove technologies

Primary stove technologies	Outcome	%	Default efficiency	Weighted Stove efficiencies
Three stone/Tripod (without air draught system/chimney)	174	66.2%	10%	7%
Basic metal wood stove	24	9.1%	20%	2%
Improved wood cookstove	25	9.5%	30%	3%
Modern Fuel Cookstove (LPG)	10	3.8%	50% ¹⁰	2%
Others (using woodstoves without grate/chimney)	30	11.4%	20%	2%
Total	263	100.0%		15.5%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

a) Baseline scenario survey

The baseline scenario survey provides critical information on target population characteristics, baseline technology use, fuel consumption, leakage, and sustainable development indicators.

The Project developer undertook a baseline scenario survey to collect the required baseline data from the target population, baseline technology use and fuel consumption, and the sustainable development indicators. The survey was conducted on the period between 07/09/2020 to 13/12/2020 by the project implementor (Greenway) and interviewed 263 households.

b) Representativeness

The baseline survey requires in-person interviews with a robust sample of end-users without project technologies that are representative of end-users targeted in the project activity.

The project implementer conducted in-person interviews with a sample representative of the end-users targeted by the project activity. The baseline surveys were conducted within the applicable geographical areas covered under the project boundary.

¹⁰ The value is based on literature findings since no default value is provided in the methodology ([Performance Of LPG Cooking Stove Using Different Design Of Burner Heads – IJERT](#))

The households surveyed were randomly selected and it was established that they were not using the project technology by the time of the survey.

The surveys were done in 10 states in India which were drawn from the six regions of India. The 10 states selected areas where the project had started distributing the stoves or where the project envisaged to distribute stoves in near future. The 10 states were found to be representative of the project boundary since it was not possible to conduct surveys in all the states in India due to the geographical spread and the Covid 19 pandemic which was limiting movements and social interactions.

c) Sample Sizing: The methodology requires that baseline survey should be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

Group size	Minimum sample size
<300	30 or population size, whichever is smaller
300 to 1000	10% of group size
> 1000	100

The targeted population for the group activity is 41,729 households. Since the group size is more than 1000, the minimum sample required shall be 100 households without using the project technology. In person interviews were held with the 263 household who were not using the project technology, to collect the required baseline data. The extra households were to cater for any sample size attritions. The sample surveyed was sufficient to fulfil the methodology requirement on sample sizing.

d) Data collected:

Information on the following was gathered during the surveys:

1. Users follow up	i. Address or location ii. Mobile telephone number and/or landline telephone number (when possible)
2. End user characteristics	iii. Number of people served by baseline technology. iv. Typical baseline technology usage patterns and tasks (commercial, institutional, domestic, etc.)

3. Baseline technology and fuels	v. Types of baseline technologies vi. Types of fuels used and estimated quantities. vii. Sources of fuels; purchased, hand-collected, or etc., and prices paid, or effort made, e.g. walking distances, persons collecting, opportunity cost viii. Fuel collection responsibilities and cooking behaviours
----------------------------------	---

e) Baseline scenario survey findings

1. End-user characteristics

The baseline survey was conducted with 263 households, who included both men and women. The average household size was calculated at 6.15.

2. Baseline fuels, stove technologies, and cooking practices

As summarised in table 5 above, 92.8% of the households were using firewood for cooking as the primary fuel, and 59.3% as secondary fuel. The main source of the baseline fuel was identified as collecting from nearby forest/wastelands (83.7%), while 9.5% of the households indicated they buy from sellers.

Table 8: Fuel acquisition by households in the baseline

Fuel acquisition	Outcome	%
Collecting from nearby forest/wastelands	220	83.7%
Buying from seller	25	9.5%
Combination of both methods	18	6.8%
Total	263	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

On fuel acquisition responsibilities, among households who were collecting fuelwood from forests, 65.8% reported it was the responsibility of women to get the fuel compared to men at 28.4%.

Table 9: Fuel collection responsibilities in the baseline

Fuel collector	Outcome	%
Children	1	0.4%
Elderly	13	5.3%
Female	160	65.8%
male	69	28.4%
	243	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

For the households who were purchasing fuel from suppliers, the duty of buying the fuel from the market was reported as 74% male and 23% female as shown in table below.

Table 10: Responsibilities of buying fuel from the market

Fuel acquisition responsibilities		
Buyer of fuel from market	Outcome	%
elderly	1	2%
female	10	23%
male	32	74%
Total	43	100%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

In terms of time spent in getting the baseline fuel, majority of the households (55.5%) reported spending more than 3 hours. Only 44.5% spent 2-3 hours in getting fuel

Table 11: Time spent in getting fuel in the baseline

Time spent getting fuel	Outcome	%
2-3 hours	117	44.5%
4-5 hours	73	27.8%
6-8 hours	63	24.0%
>8 hours	10	3.8%
Total	263	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

On cooking behaviours, women were observed to be the primary cook in the households as reported by 98.9% as summarised below.

Table 12 Cooking responsibilities in households

Primary cook in households	Outcome	%
Female	260	98.9%
Male	2	0.8%
Children	1	0.4%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

The time spent on cooking using the baseline fuels was also assessed where majority (56.7%) spent between 1-2 hours. 3.4% spent more than 2 hours, while 8.4% spent less than 45 minutes. The average number of meals cooked per day was calculated at 2.4.

Table 13: Time spent cooking

Time spent cooking	Outcome	%
<45 minutes	22	8.4%
45-60 min	83	31.6%
1-2 hours	149	56.7%
>2hours	9	3.4%
Total	263	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

3. Baseline Sustainable development assessment

The survey respondents were asked to rate the current level of indoor smoke within their households. A majority (75.3%) expressed the smoke level range was between moderate to very high while the rest (20.5%) expressed the level as between low to no smoke at all.

Table 14: Smoke levels in households

Smoke levels	Outcome	%
Very high	23	8.7%
High	83	31.6%
Moderate	92	35.0%
Low	16	6.1%
Very low	31	11.8%
No smoke	7	2.7%
No answer	11	4.2%
	263	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

The respondents were also asked about the hospital visit frequency for the person using the traditional cookstoves due to smoke related problem in a year. Only 3.4% of the respondents had not visited the hospital, with majority (78.3%) having visited the hospital at least twice in a year.

Table 15: Frequency of hospital visits due to smoke related problem

Hospital visits	Outcome	%
no visit	9	3.4%
1-2 times	206	78.3%
3-4 times	46	17.5%
5-6 times	1	0.4%
More than 7	1	0.4%
	263	100.0%

Source: Baseline survey data spreadsheet tab 2020-baseline survey

The project activity is a non-industrial activity and only a single technology is installed in a household. The installation is done gradually. In line with the applicable methodology, the baseline is fixed until the end of the registered crediting period and will not require continuous monitoring or updating. If a project cookstove is replaced with a cookstove of similar efficiency prior to the end of the crediting period, the original baseline shall be applicable till the end of the replaced cookstoves' useful lifetime or the registered crediting period, whichever occurs earlier.

The proposed VPA, will disseminate Greenway Smart, Greenway Prime, and Greenway Jumbo cookstoves with a thermal efficiency of 32.098%, 29.8%, and 38%, respectively which will replace the baseline stoves currently used for domestic cooking by the target communities. This action will lead to a reduction in the use of non-renewable biomass, and this will be the new project scenario. In the project scenario, highly efficient stoves are used in households, which consume less non-renewable biomass compared to the baseline scenario.

4. Baseline non-renewable biomass (NRB) assessment

The baseline scenario identified by the project activity assumes use of non-renewable biomass (firewood and charcoal) as the main baseline fuel among the targeted population. The fractional non-renewability of biomass has been established using the latest CDM tool, "Tool 30: Calculation of the fraction of non-renewable biomass, version 3.0". The value of f_{NRB} of 0.8013 for India has been calculated and this is the value used in this VPA.

The value of f_{NRB} is fixed ex-ante for the entire crediting period.

Determination of the Share of Non-Renewable Biomass

According to methodological TOOL 30 vers.3.0, par. 7, the following equation shall be used to calculate f_{NRB} :

$$f_{NRB,y} = \frac{NRB}{NRB+RB} \quad (\text{Equation 1})$$

where:

$f_{NRB,y}$	Fraction of non-renewable biomass in the country (fraction or %)
NRB	Quantity of non-renewable biomass consumed (t/yr)
RB	Quantity of renewable biomass (t/yr)

$$NRB = H - RB$$

Where:

H	=	Total consumption of woody biomass in the applicable area in the relevant period (tonnes), determined either through: (a) Official statistics or reports or peer-reviewed literature;
---	---	---

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other}))$$

Where	
$MAI_{forest,i}$	Mean Annual Increment of woody biomass growth per hectare in sub-category of forest areas (t/ha/yr)
$MAI_{other,i}$	Mean Annual Increment of woody biomass growth per hectare in sub-category of other wooded land areas (t/ha/yr)
$F_{forest,i}$	Extent of forest in sub-category I (ha)
$F_{other,i}$	Extent of other wooded land in sub-category I (ha)
P_{forest}	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha)
P_{other}	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha)
i	Sub-category i of forest areas and other wooded land areas

Determination of H			
Type of Wood	Unit	Wood consumption	Source
Fuelwood	tons/y	85,289,795	https://fsi.nic.in/isfr-volume-ii?pgID=isfr-volume-ii
Fodder	tons/y	1,053,038,714	https://fsi.nic.in/isfr-volume-ii?pgID=isfr-volume-ii
Bamboo	tons/y	21,954,832	https://fsi.nic.in/isfr-volume-ii?pgID=isfr-volume-ii
Small timber Volume	cum/	5,848,202	https://fsi.nic.in/isfr-volume-ii?pgID=isfr-volume-ii
Wood Density "Shorea robusta" in t/cum	t / cum	1	2006 IPCC
Small timber Mass	tons/y	4,211	Calculated
Total H India	t / yr	1,160,287,552	

Calculation of RB		Tropical					
	Unit	Rain Forest	Moist	Dry	Shrub	Mountain	Subtropical Mountain
$F_{forest,i}$	%	13%	11%	56%	9%	7%	5%
$MAI_{forest,i}^{11}$	(t/ha/yr)	3.4	2.4	3.9	5	2.9	2.5
$MAI_{forest,i} * (F_{forest,i} - P_{forest})$	ha	27,784,827	16,595,462	137,289,734	28,287,720	12,760,905	7,857,700

RB	t/yr	230,576,349
H	t/yr	1,160,287,552
NRB	t/yr	929,711,203
f_NRB		0.8013

5. Baseline firewood consumption and target population characteristics

The quantity of firewood consumed in the baseline has been determined based on the estimated average annual consumption of firewood per household (tonnes/year) established through a baseline survey carried out by the project.

Due to the COVID-19 pandemic, the VPA did not carry out a KPT on time before the validation as required by the methodology. The VPA has therefore applied the survey data as proxy values for purposes of ex-ante emission reduction calculation, pending the results of the KPT that will be carried out by the project before the 1st verification. The Kitchen

¹¹ Updated values sourced from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 4.9 (https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf)

Performance Test Procedure stipulated under Annex 2 of the applicable methodology shall be applied.

The VPA had carried out a baseline survey whereby estimated fuel consumptions were provided by the target end users. These are the values which have been used to estimate the ex-ante emission reductions. However, a KPT will be carried out before the first verification to determine both fuel consumption and savings. The results will be applied in the emission reduction calculations.

The baseline fuel consumption is shown in the table below:

Baseline stove	Fuel consumed (kg/day)
Chulas	7.93

Therefore, the quantity of fuel consumed in the baseline ($B_{b,y}$) is determined as 7.9 kg/day (See ER spreadsheet, tab "Analysis-2020").

Using the below formula, the project fuel savings for each technology was calculated as shown below:

Jumbo stoves

$$Fuel_{baseline} = \eta_{project} / \eta_{baseline} \times Fuel_{project}.$$

$Fuel_{baseline}$	7.93
$\eta_{project}$	38%
$\eta_{baseline}$	15.5%
$Fuel_{project}$	3.23
$Fuel_{savings}$	4.69

Smart stoves

$$Fuel_{baseline} = \eta_{project} / \eta_{baseline} \times Fuel_{project}.$$

$Fuel_{baseline}$	7.93
$\eta_{project}$	32.10%
$\eta_{baseline}$	15.5%
$Fuel_{project}$	3.83
$Fuel_{savings}$	4.10

Prime stoves

$$Fuel_{baseline} = \eta_{project} / \eta_{baseline} \times Fuel_{project}.$$

$Fuel_{baseline}$	7.93
$\eta_{project}$	29.80%
$\eta_{baseline}$	15.5%
$Fuel_{project}$	4.12
$Fuel_{savings}$	3.80

6. Relevant applicable legislation and how effectively these are enforced (GS4GG Principle 1).

In India, there is no requirement that households must purchase only efficient cookstove or any project which targets to implement a domestic household project must install efficient cookstoves. This, therefore, is a voluntary action by the project developer to initiate this project whereby households will be provided with an efficient cook stove at a flexible repayment method. The credit facility and related flexibility of payment makes it possible for majority of the end users to have access to the stove. Therefore, the project is not being implemented as a result of any existing regulatory requirements. In addition, the ministry of New and Renewable Energy provides a list of all approved improved cookstoves which can be distributed in the country and the Greenway stoves are among those stoves which have been listed as approved by the ministry.

B.5. Demonstration of additionality

Section 3.3.1 of the applied methodology, TPDDTEC Version 4.0 provides that "The project developer must show that the project could not or would not take place without the presence of carbon finance".

Further, the project developer shall demonstrate additionality prior to project registration by conforming to additionality requirements of one of the options below,

- a. Applicable GS4GG Activity Requirements.
- b. CDM Tool 01 - Tool for the Demonstration and Assessment of Additionality.
- c. CDM Tool 19- Demonstration of additionality of microscale project activities; (not applicable to Gold Standard microscale projects)
- d. CDM Tool 21 – Demonstration of additionality of small-scale project activities; (applicable to small-scale projects only)
- e. An approved Gold Standard VER additionality tool

Pursuant to section 4.1.9 of Community activity service requirements, "Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification".

(a) Positive list

(b) Projects located in LDC, SIDS, LLDC4

(c) Microscale projects

The Community Services Activity Requirements V1.2, Annex B on positive list, section 1.1.5 lists "*Project activities that involve technologies and/or practices providing thermal energy to the user that have less than 20% adoption rate among the target users*" as additional. The section further requires that the most recent available data on adoption rates shall be used to demonstrate compliance with the 20 per cent threshold in the

target geographical area. In no case shall data be used if older than three years from the date of commencement of validation of the project activity. The additionality of the project has been established using this provision.

The project activity entails distribution of improved firewood cookstoves to deliver thermal energy to households for domestic cooking. The primary targets are end-users using inefficient biomass stoves especially those in rural areas.

According to the *State of Clean Cooking Energy Access in India* report by CEEW (2020)¹², half of Indian households continue to use firewood, and nearly a fourth of the households reported using dung cakes and other solid fuels for cooking despite improvements in LPG use. Most solid fuel users are in rural India, where 67% of the households use firewood for cooking. This practice is a concern due to households continued high exposure to Household Air Pollution (HAP). The *Roadmap for Access to Clean Cooking Energy in India* report by the Council on Energy, Environment and Water (CEEW) published in 2019¹³ reports that, less than 1% of rural households surveyed reported using improved biomass cookstoves (ICS). Only 14% of households surveyed in the report were aware of their existence, indicating low awareness among non-users (refer to section 5.3 on page 38).

The Government of India have established policies in the past to increase the penetration rate of ICS. One of such programmes is the National Programme on Improved Chulhas (NPIC), which sought to introduce 35 million chulhas between 1986 and 2002 (MNES, 2004). However, the Global Alliance for Clean Cookstoves, India Cookstoves and Fuels Market Assessment report (2013) showed that only 4% of Indian households used ICS by 2013. Therefore, the program did not achieve the intended objective by 2013.

The other program introduced by the Government in 2014 was the Unnat Chulha Abhiyan (UCA), with the aim of deploying 2.75 million ICS by March 2017 (MNRE, 2014b). However, the report by Council on Energy, Environment and Water (CEEW) in 2019 indicate that the scheme had met only 1.3 per cent of its target by March 2017, with much of the budget having lapsed unutilised.

The findings from the baseline survey conducted by the project also indicated that only about 9.5% of the respondents were using improved firewood cookstoves.

¹² [ires-report-on-state-of-clean-cooking-energy-access-in-india.pdf \(ceew.in\)](https://www.ceew.in/reports/state-of-clean-cooking-energy-access-in-india)

¹³ [ResearchGate](https://www.researchgate.net/publication/354111111)

Based on the above discussions, it is evident that significantly less than 20% of Indian households that use biomass for cooking use non-improved biomass cookstoves. The additionally criteria set out in section 1.1.5 is therefore met and the project activity is deemed additional.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects
Describe how the proposed project meets the criteria for deemed additionality.	The project meets criterion 1. Positive list, Annex B of the Community Services Activity Requirements. Section 1.1.5 of Annex B lists "Project activities that involve technologies and/or practices providing thermal energy to the user that have less than 20% adoption rate among the target users" as additional. As discussed above, the adoption rate of improved biomass stoves by 2019 was less than 20% and hence the project is additional.

B.5.1. Prior Consideration

According to the Gold Standard's Principles and Requirements (version 1.2), Retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.

The first stakeholder meeting was held on 14/10/2020 and the second was held on 29/01/2021. The stove distribution under the project activity started on 29/10/2020 which was after the first consultation. The project cycle has therefore been identified as regular and the project is hence not required to demonstrate prior consideration. The

first submission to the GS was made on 19/03/2021 when the LSC documentation was uploaded under the PoA for design consultation review, which is within 1 year from the start of the project.

The project is eligible for Gold Standard certification and the key milestones are as summarized below.

Table 16: Key project milestones

Timeline	Milestone
14/10/2020	The first LSC meeting
29/10/2020	Start date of the project (first unit distribution).
29/01/2021	Second LSC meeting
19/03/2021	The first submission date to GS
08/07/2021 - 08/09/2021	Stakeholder feedback round (SFR)
10/06/2021	PoA Design review closed
06/07/2021	VVB contacting for validation
09/03/2022	Project listing

B.5.2. Ongoing Financial Need

Investment in stoves production, distribution and sale requires a lot of investment and when the target market is made up of mostly low-income earners, an appropriate credit facility with an appropriate repayment plan is required to enable most users to purchase the stove. The repayment plans, such as a staggered repayment plan, drain the capital available for running the business. Without an alternative revenue generation, the financial sustainability and/or expansion of the project of the project is therefore compromised, with most people not being reached to benefit from the project. The project currently sells the following stoves:

- Smart stove for Indian Rupees, ₹1,799.00 (USD23.10)
- Jumbo stoves for ₹2,999.00 (USD38.51)
- Prime stoves for ₹3,299(USD42.37).

These prices have been subsidised by up to 30% by the carbon revenues. Comparing the current stove price with average household income levels in India where the average monthly earning is ₹16,840 with an expenditure of ₹11,101, leading to a monthly

disposable income of ₹5,739¹⁴, the cost of the stove is about a half of the monthly disposable income for an average Indian family. Without the carbon revenues contribution, the price of the stove would increase by 30% and this will make them more costly to average users.

This project activity is funded by carbon revenues in advance and the project would not have occurred without the advance carbon revenues provided to the project. since the project is made possible by the carbon revenue, the proposed VPA, therefore, is additional.

Carbon revenues will contribute to the sustainability of the proposed VPA by making stoves affordable to more potential customers in several states of India. The need for ongoing financial support helps in making the project sustainable.

¹⁴ <https://www.livemint.com/Politics/AvHvyHVJIhR0Q629wkPS5M/Indias-richest-20-account-for-45-of-income.html>

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action	13.2 Integrate climate change measures into national policies, strategies, and planning.	Amount of Verified Emission Reductions (VERs)
1. No Poverty	1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day	Estimated monetary savings computed from fuel saved
5. Gender Equality	5.4: Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Average time saved by women in fuel collection
7. Access to clean energy	7.1: By 2030, ensure universal access to affordable, reliable and modern energy services	Number of people/households with access to basic service provided by the project device
8. Decent Work and economic growth	8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Total number of jobs split by gender

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

Methodological choices/approaches related to SDG 13

Since the baseline fuel and project fuel are the same, emission reductions are exclusively from improved efficiency. The project, therefore, shall use Method 1 in calculating the emission reduction. Therefore, the GHG emissions reduction achieved by the project activity in year y is calculated using equation 1. Equation one has been adjusted to include leakage where the project has applied a default value of *0.95 to equation 1. In line with the applied methodology TPDDTEC, version 4.0, emission reductions can be estimated by using the following formulas.

$$ER_y = \sum_{b,p} (N_{b,p,y} \times U_{p,y} \times SFS_{p,b,y} \times NCV_{b,fuel} \times (f_{NRB,b,y} \times EF_{b,f,CO2} + EF_{b,f,nonCO2})) * \sum LE_{p,y} \quad Eq. 1$$

Where:

ER_y	Emission reduction for total project activity in year y (tCO ₂ e/yr)
$\sum_{b,p}$	Sum over all relevant baseline b/project p pairs
$N_{b,p,y}$	Number of project technology-days included in the project database for baseline b/project p pair in year y (days)
$U_{p,y}$	Cumulative Usage rate for technologies in project scenario p in year y (fraction)
$SFS_{p,b,y}$	Specific fuel savings for an individual project technology of baseline b/project p pair in year y (mass or volume units/technology*day)
$NCV_{b,fuel}$	Net calorific value of the fuel(s) that is substituted or reduced in baseline b (TJ/mass or volume units)
$f_{NRB,b,y}$	Fractional non-renewability status of woody biomass fuel during year y (fraction). For biomass, it is the fraction of woody biomass that can be established as non-renewable. This parameter is omitted when f is a fossil fuel.
$EF_{b,f,CO2}$	CO ₂ emission factor from use of fuel f (tCO ₂ /TJ)
$EF_{b,f,nonCO2}$	Non-CO ₂ emission factor arising from use of fuel f, when the baseline fuel f is biomass or charcoal (tCO ₂ e/TJ). This parameter is omitted when f is a fossil fuel.
$LE_{p,y}$	Leakage for project scenario p in year y (tCO ₂ e/yr)

Parameter	Description	Value applied	Unit	Data source
$\Sigma_{b,p}$	Sum over all relevant (baseline b/project p) couples	1	Number	Internal assumption for ex-ante purposes
$N_{p,y}$	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y	1	Number	Internal assumption for ex-ante purposes
$U_{p,y}$	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction)	1	Fraction	Internal assumption for ex-ante purposes
$SFS_{p,b,y}$	Specific fuel savings for an individual project technology of baseline b/project p pair in year y (mass or volume units/technology*day)	1.71 (Jumbo) 1.50 (smart) 1.39 (prime)	tons/year	Calculated using baseline estimates
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)	0.8013	Fraction	Calculated
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)	0.0156	TJ/ton	Default value as per applied methodology
$EF_{b,fuel,CO_2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel	112	tCO ₂ /TJ	Default value as per applied methodology
$EF_{b,fuel,nonCO_2}$	CO ₂ emission factor arising from use of fuels in project scenario	8.692	tCO ₂ /TJ	Default value as per applied methodology (AR4 GWP)
$NCV_{p,fuel}$	Net calorific value of the fuels used in the project	0.0156	TJ/ton	Default value as per applied methodology
$EF_{p,fuel,CO_2}$	Non-CO ₂ emission factor arising from use of fuels in project scenario	112	tCO ₂ /TJ	Default value as per applied methodology
$EF_{p,fuel,nonCO_2}$	Non-CO ₂ emission factor of the fuel that is reduced	8.692	tCO ₂ /TJ	Default value as per applied methodology (AR4 GWP)
$LE_{p,y}$	Leakage for project scenario p in year y	0.95	tCO ₂ e/yr	Internal assumption for ex-ante purposes

ER _y	Emission reduction for total project activity in year y	2.41 (Jumbo) 2.08 (smart) 1.91 (prime)	tCO2 e/yr	Calculated
-----------------	---	---	--------------	------------

Source: GS11358 ER calculations spreadsheet tab ER calculations

Determination of Specific fuel savings (SFS_{p,b,y})

The Specific fuel savings for an individual project technology of baseline b/project p pair in year y(SFS_{p,b,y}) is determined through proxy data since the project did not carry out actual KT due to the COVID-19 pandemic, instead, it applied estimated fuel consumption values obtained from the baseline survey which the project carried out before the pandemic. However, prior to the first verification, the project will carry out a KPT to get new values which will be updated in the PDD and used in the monitoring report and emission reduction calculations.

The specific fuel savings was calculated as shown below:

Jumbo stoves Parameter	Value
Baseline stove efficiency	15.5%
Project stove efficiency	38.0%
Fuel use per year (tonnes)	7.93
Fuel use per day (kg)	2.89
	Daily fuel consumption (kg)
Fuel _{baseline}	7.93
n _{project}	38.0%
n _{baseline}	15.5%
Fuel _{project}	3.23
Fuel _{savings}	4.69
Baseline fuel consumption Fuel _{baseline} (tonnes/yr)	2.89
Yearly fuel savings, Fuel_{savings} (tonnes)	1.71

Source: GS11358 ER spreadsheet, tab specific savings

Smart stoves

Parameter	Value
Baseline stove efficiency	15.5%
Project stove efficiency	32.10%
Fuel use per year (tonnes)	7.93
Fuel use per day (kg)	2.89
	Daily fuel consumption (kg)
Fuel _{baseline}	7.93
n _{project}	32.10%
n _{baseline}	15.5%
Fuel _{project}	3.83
Fuel _{savings}	4.10
Baseline fuel consumption Fuel _{baseline} (tonnes/yr)	2.89
Yearly fuel savings, Fuel_{savings} (tonnes)	1.50

Source: GS11358 ER spreadsheet, tab specific savings

Prime stoves

Parameter	Value
Baseline stove efficiency	15.5%
Project stove efficiency	29.80%
Fuel use per year (tonnes)	7.93
Fuel use per day (kg)	2.89
	Daily fuel consumption (kg)
Fuel _{baseline}	7.93
n _{project}	29.80%
n _{baseline}	15.5%
Fuel _{project}	4.12
Fuel _{savings}	3.80
Baseline fuel consumption Fuel _{baseline} (tonnes/yr)	2.89
Yearly fuel savings, Fuel_{savings} (tonnes)	1.39

Source: GS11358 ER spreadsheet, tab specific savings

Leakage

The leakage emissions, $LE_{p,y}$, for the project is determined using Option 1, whereby a default adjustment factor of 0.95 to the emission reductions to approximate leakage emissions is used in the ER estimation. Therefore, the term " $-\sum LE_{p,y}$ " in equations equals to " $* 95\%$ ".

Methodological choices/approaches related to SDG 1

The VPA includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cooking methods and thus result in a reduction in firewood consumption. The provision of the technology a flexible repayment method coupled with the reduced fuel demand for the daily cooking activities increases access to affordable energy and improved cooking technologies (i.e. basic services) for the stove users. This also enables households to spend less on fuel purchases, therefore, making savings.

The contribution of the project to this target of SDG 1 will be measured by calculating the estimated household savings based on the prevailing fuel prices. This will be calculated as follows:

$$\text{Net Benefit (SDG 1)} = \text{Financial savings}_{\text{baseline}} - \text{Financial savings}_{\text{project}}$$

The project is expected to lead to saving of about 21.51 Rupee/household/day on fuel costs.

Methodological choices/approaches related to SDG 5

The use of a Greenway stove in a household helps in reducing the time spent on firewood collection which is mostly done by women. Fuel collection takes more time and effort, which takes away productive hours of women and young girls leading to time poverty and lost opportunity.

The average time saved by women during fuel collection will be determined as follows:
= Average fuel collection time = the average number of times women in a household used to fetch firewood in the baseline scenario – the average number of times women in a household fetch firewood in the project scenario.

In the baseline, 55.5% of the households spent between 4-8 hours per day in fuel collection. By using the project technology, the number of hours spent per household in fuel collection is expected to reduce by at least 3.45 hours per day.

Methodological choices/approaches related to SDG 7

Parameter: Number of people/households with access to basic service provided by the project device

This parameter will be monitored based on the actual number of operational units sold as indicated in the sales database. The project implementor shall maintain an updated sale database for all the units sold by the project.

The project is expected to distribute about 41,729 improved fuelwood cookstoves among household end-users.

Methodological choices/approaches related to SDG 8

The VPA is expected to job creation through the stove production and distribution process. This parameter will be monitored through project reports on the number of people employed. The reporting will also include data on the number of males and number of females involved in the project.

In the baseline there were no people employed by the project. In the project scenario, the project targets to employ 300 people including 200 male and 100 female in the stove production and supply chain.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter ID	SDG 13
	ICS 1
Data/parameter	Baseline scenario survey results
Unit	N/A
Description	Report of the results of the baseline scenario survey
Source of data	The report presents the results of the Baseline Scenario Survey, relevant for the baseline scenario definition.
Value(s) applied	The baseline scenario findings are presented in section B.4 of the VPA-DD and the values are presented on tables 4 to 15.
Choice of data or Measurement methods and procedures	The applied data have been informed by the findings of the baseline survey.
Purpose of data	Calculation of baseline emissions
Additional comment	A baseline survey was undertaken as detailed in the VPA-DD section B.4. The surveys were done with 263 participants who were not using the project technology. The survey was undertaken between 07/09/2020 to 13/12/2020 in 10 states which included Assam, Bihar, Chhattisgarh, Gujarat, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Tamil Nadu and Uttar Pradesh. The findings obtained from the survey have been reviewed and have been found to be representative of the baseline scenario. The data will be applied during the crediting period.

Data/parameter ID	SDG 13																																															
	ICS 2																																															
Data/parameter	Project technology description																																															
Unit	NA																																															
Description	The detailed description of the project technology																																															
Source of data	The methodology requires that any of the following sources shall be used: - Manufacturer specifications - Certifications by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee - Technical reports from the installer - For stoves built on-site at the end user location, reports of Standard WBT by stove manufacturer or installer Professional opinion or expert opinion is not accepted as a source for this parameter. <i>The Project technology description has been made based on manufacturer specifications.</i>																																															
Value(s) applied	<table><tr><th>Parameters</th><th>Smart Stove</th><th>Jumbo Stove</th><th>Prime stoves</th></tr><tr><td>Manufacturer name</td><td>Greenway Grameen Infra pvt</td><td>Greenway Grameen Infra pvt</td><td>Greenway Grameen Infra pvt</td></tr><tr><td>Size:</td><td>9.8" x 7.6" x 11.6"</td><td>12.4" x 10.6" x 11.6"</td><td>10" x 9" x 11.9"</td></tr><tr><td>Materials:</td><td>Steel and Aluminium with Bakelite Handles</td><td>Steel and Aluminium with Bakelite Handles</td><td>Steel and Aluminium with Bakelite Handles</td></tr><tr><td>Loading Capacity:</td><td>25 kg</td><td>40 kg</td><td>40kg</td></tr><tr><td>Secondary Air Induction Mechanism:</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Warranty:</td><td>1 year</td><td>2 years</td><td>2 years</td></tr><tr><td>Fuel Type</td><td>Ergonomic front loading design</td><td>Ergonomic front loading design</td><td>Ergonomic front loading design</td></tr><tr><td>Operational Lifetime (lifespan)</td><td>72 months (6 years)</td><td>72 months (6 years)</td><td>72 months (6 years)</td></tr><tr><td>Fuel type</td><td>Fuelwood</td><td>Fuelwood</td><td>Fuelwood</td></tr><tr><td>Application/ service level</td><td>Domestic cooking</td><td>Domestic cooking</td><td>Domestic cooking</td></tr></table>				Parameters	Smart Stove	Jumbo Stove	Prime stoves	Manufacturer name	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt	Size:	9.8" x 7.6" x 11.6"	12.4" x 10.6" x 11.6"	10" x 9" x 11.9"	Materials:	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles	Loading Capacity:	25 kg	40 kg	40kg	Secondary Air Induction Mechanism:	Yes	Yes	Yes	Warranty:	1 year	2 years	2 years	Fuel Type	Ergonomic front loading design	Ergonomic front loading design	Ergonomic front loading design	Operational Lifetime (lifespan)	72 months (6 years)	72 months (6 years)	72 months (6 years)	Fuel type	Fuelwood	Fuelwood	Fuelwood	Application/ service level	Domestic cooking	Domestic cooking	Domestic cooking
Parameters	Smart Stove	Jumbo Stove	Prime stoves																																													
Manufacturer name	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt	Greenway Grameen Infra pvt																																													
Size:	9.8" x 7.6" x 11.6"	12.4" x 10.6" x 11.6"	10" x 9" x 11.9"																																													
Materials:	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles	Steel and Aluminium with Bakelite Handles																																													
Loading Capacity:	25 kg	40 kg	40kg																																													
Secondary Air Induction Mechanism:	Yes	Yes	Yes																																													
Warranty:	1 year	2 years	2 years																																													
Fuel Type	Ergonomic front loading design	Ergonomic front loading design	Ergonomic front loading design																																													
Operational Lifetime (lifespan)	72 months (6 years)	72 months (6 years)	72 months (6 years)																																													
Fuel type	Fuelwood	Fuelwood	Fuelwood																																													
Application/ service level	Domestic cooking	Domestic cooking	Domestic cooking																																													

	<table><tr><td>Energy output</td><td>1.34 KW</td><td>2.32 KW</td><td>TBC¹⁵</td></tr><tr><td>Efficiency</td><td>32.098%</td><td>38%</td><td>29.8%</td></tr></table>	Energy output	1.34 KW	2.32 KW	TBC ¹⁵	Efficiency	32.098%	38%	29.8%
Energy output	1.34 KW	2.32 KW	TBC ¹⁵						
Efficiency	32.098%	38%	29.8%						
Choice of data or Measurement methods and procedures	Greenway Grameen Infra private is the manufacturer of the project technology, and the manufacturers specifications provide the detailed description of the product technology.								
Purpose of data	Calculation of baseline emissions								
Additional comment	For any information not available at the time of validation, validating VVB shall include a FAR.								

¹⁵ To be determined before 1st verification

Data/parameter ID	SDG 13
	ICS 3
Data/parameter	Expected technical life of project technology
Unit	years
Description	The expected technical life of an individual project technology shall be defined in the VPA-DD.
Source of data	The methodology provides that any of the following sources shall be used: - Manufacturer specifications - Certification by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee or Guarantee from the installer - For stoves built on-site at the end user location, field reports, which comply with the general requirements for sampling, of average technical life of the same stove type operated under similar conditions (socioeconomic and cultural). Simulation modelling may be applied together with such field reports to estimate the average technical life. Professional opinion or expert opinion is not accepted as a source for this parameter. The PP has sourced the data for the parameter from manufacturer guarantee as per the provided confirmation letter.
Value(s) applied	6
Choice of data or Measurement methods and procedures	Greenway Grameen Infra pvt is the manufacturer of the project technology, and the letter of guarantee serves as a confirmation of the expected technical life of the project technology. Greenway stoves customers have reported unencumbered use of the stoves without any crucial functionality of the stove hindered due to any manufacturing faults. Greenway has stringent quality control checks at their manufacturing facility to ensure that the customers can use the stove with minimal wear and tear affecting the performance and efficiency of the stoves for 6 years in ordinary conditions.
Purpose of data	Calculation of baseline emissions
Additional comment	The expected technical life of project technology is longer than the 5-year crediting period of the project and therefore measures to ensure that end users are provided replacement technology of comparable or higher quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee is not needed.

Data/parameter ID	SDG 3
	ICS 4
Data/parameter	Indoor air pollution (IAP) levels of the project technology
Unit	NA
Description	For projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (for example, changing from a stove with a chimney to improved stove with no chimney), demonstration that Indoor air pollution (IAP) levels are not worsened in the project scenario compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions.
Source of data	The methodology provides that for the description of IAP level of the project technology, any of the following sources shall be used: - Certification resulting from a manufacturer's test - Report of field testing of the technology - Report of lab testing of the technology - Results of modelling of the technology's operation under field conditions. - For stoves built on-site at the end user location, existing reports of lab or field testing of similar technology. Professional opinion or expert opinion is not accepted as a source for this parameter. For the IAP level of the baseline scenario, the following sources shall be used: - Certification resulting from of a manufacturer's test, - Report of field testing of the technology, - Report of lab testing of the technology, - Results of modelling of the technology's operation under field conditions, or - Expert opinion For both project and baseline technologies, if none of these are available, reference from published literature or report by independent agencies may be used as evidence, provided it is not more than 5 years old. For the description of IAP level of the project technology, a report of lab testing of the technology is used. For the IAP level of the baseline scenario, reference from published literature reports by independent agencies shall be used as evidence, provided it is not more than 5 years old.
Value(s) applied	Project IAP – WBT test report values:

	Jumbo: 394.49¹⁶(PM 2.5 mg/MJd) Smart: 1.6 (Total Suspended Particulate Matter (mg/mt), CO/CO2 ratio: 0.0265) Prime: 328.79¹⁷(PM 2.5 mg/MJd)) Baseline IAP: 695¹⁸ 24-hr µg/m3
Choice of data or Measurement methods and procedures	WBT test report for project IAP and Literature review of published data for baseline IAP
Purpose of data	Calculation of baseline emissions
Additional comment	A study published by the Clean Cooking Alliance in 2016 on "Analysing the costs and benefits of clean and improved cooking solutions (conducted by Sanford School of Public Policy and Duke Global Health Institute) shows that the PM2.5 concentrations from wood burning using traditional stoves was estimated at 695 24-hr µg/m3. Another study done in 2017 on "Comparison of a Traditional Cook Stove with Improved Cook Stoves Based on Their Emission Characteristics", reports a PM2.5 value of 680¹⁹ µg/m3+/-128. The first value has been used for purposes of the ex-ante estimation for PM2.5 of the baseline stoves since its more conservative.

¹⁶ Medium value was 305.97 and CO produced was 4.64g/MJd

¹⁷ High power

¹⁸ [355-1.pdf \(cleancooking.org\)](#) see table 5 on page 19

¹⁹ [\(PDF\) Comparison of a Traditional Cook Stove with Improved Cook Stoves Based on Their Emission Characteristics \(researchgate.net\)](#) see table 1 on page 2

Data/parameter ID	SDG 13 ICS 5
Data/parameter	Avoidance of double counting or double claiming among project participants
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with other parties directly involved with the project or programme.
Source of data	Written assertions with the project developer of the ownership rights and intention of selling the emission reductions resulting from the project activity directed at or signed with all the applicable parties of the following: - Carbon rights waiver signed by end-user transferring carbon rights to Greenway Grameen Infra Private Limited - Emission Reduction Purchase Agreement signed between Greenway Grameen Infra Private Limited transferring the carbon rights to ClimateCare Oxford Limited.
Value(s) applied	The PP has provided proof that the procedure for the carbon rights transfer with end-users of the project technology is in place.
Choice of data or Measurement methods and procedures	The mechanism for the carbon rights transfer has been put in place and the same shall be verified to ensure there is no double counting or claiming by the project participants.
Purpose of data	Calculation of baseline emissions
Additional comment	Any written assertions not available at validation shall be included as a FAR and be provided and verified at the time of first verification

Data/parameter ID	SDG 13 ICS 6
Data/parameter	Avoidance of double counting or double claiming with other mitigation actions
Unit	NA
Description	Review and analysis of mitigation actions in other voluntary market or UNFCCC/compliance mechanisms
Source of data	Using publicly available information from Gold Standard and other voluntary standards, at a minimum Verra and any recognized national or regional standards in the project location, and UNFCCC CDM project & PoA database: - Identify and list any mitigation actions of similar technology, i.e., that provide the same kind of output and use the same kind of equipment or conversion process, operating in overlapping spatial boundaries. If one or more are identified: - Describe the practices that will be implemented to ensure that the programme or project activity quantifies emission reductions only from technology it has implemented, - Describe the practices to avoid that the programme or project activity implementation displaces technology of other mitigation actions, and - Design a method to discount emission reductions in case the programme or project activity is found to displace or operate alongside another mitigation action. The PP has reviewed the Gold Standard and, Verra and UNFCCC CDM project & PoA registries
Value(s) applied	GS10818: Dissemination of Improved Cookstoves in India by Greenway - PoA 1. GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway - VPA001 2. GS10818 - Dissemination of Improved Cookstoves in India by Greenway - Dissemination of Improved Cookstoves in Karnataka by Greenway - VPA002 3. GS11242- Fair Climate Programme for Sustainable Household Energy (PoA) <u>CDM 9181: MicroEnergy Credits – Microfinance for Clean Energy Product Lines – India (unfccc.int)- POA</u> 1. MicroEnergy Credits POA – Grameen Koota Indoor Air Pollution Campaign 2. MicroEnergy Credits PoA – CPA 02 3. MicroEnergy Credits PoA - CPA 03 4. MicroEnergy Credits PoA - CPA 04 5. MicroEnergy Credits PoA - CPA 06 6. MicroEnergy Credits PoA - CPA 07 7. MicroEnergy Credits PoA - CPA 08

	8. MicroEnergy Credits PoA – CPA-19- Clear Sky Partners 9. MicroEnergy Credits PoA - CPA 11 10. MicroEnergy Credits PoA – CPA 12 11. MicroEnergy Credits PoA – CPA 13 12. MicroEnergy Credits PoA – CPA 15 13. MicroEnergy Credits PoA – CPA 16 14. MicroEnergy Credits PoA – CPA 17 15. MicroEnergy Credits PoA – CPA 18 16. MicroEnergy Credits PoA – CPA-21- Clear Sky Partners 17. MicroEnergy Credits PoA – CPA-24- Clear Sky Partners 18. MicroEnergy Credits PoA – CPA-26- Clear Sky Partners 19. MicroEnergy Credits PoA – CPA-29- Clear Sky Partners 20. MicroEnergy Credits PoA – CPA 31 21. MicroEnergy Credits PoA – CPA 35 22. MicroEnergy Credits PoA – CPA 36 23. MicroEnergy Credits PoA – CPA 37
Choice of data or Measurement methods and procedures	Data is sourced from the official project registries of the carbon standards.
Purpose of data	Calculation of baseline emissions
Additional comment	Greenway Grameen Infra Private Limited is implementing another Gold Standard Project which uses their project stoves, in addition, the Greenway stoves are also used in another CDM registered project. The following practices will be in put in place to avoid any potential double-counting: 1. – In order to ensure that the project activity quantifies emission reductions only from technology it has implemented, each project activity shall maintain a separate and distinct end-user database. 2. Each stove shall be registered under only one project activity and will be clearly distinguished by a unique serial number and cannot be included in any other project. 3. To avoid that the project activity implementation displaces technology of other mitigation actions, a question will be included in the monitoring survey questionnaire to cross-check if the end-users are using other stoves included in other programmes, and the frequency of use. 4. In case the project technology is found to displace or operate alongside other similar stove technologies included in other carbon standards, the stove will be discounted in the emission reductions.

Data/parameter ID	SDG 13 ICS 7
Data/parameter	Regulatory framework for provision of thermal energy services
Unit	NA
Description	Evidence that the project does not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply/devices or fuel supply or use
Source of data	List and provide a summary of any national, sub-national and local regulations or guidance for provision of thermal energy services/devices of the type of the project provides in the project boundary, including any tariff requirements. Describe how the project complies with the regulatory framework.
Value(s) applied	See table 17 below
Choice of data or Measurement methods and procedures	The PP reviewed the regulatory framework for provision of thermal energy services within the host country (India). From the assessment, it has been demonstrated that the project activity does not undermine nor conflict with any regulation or guidance relating to thermal energy supply/devices or fuel supply or use. A detailed analysis is provided in Table 17 below.
Purpose of data	Calculation of baseline emissions
Additional comment	Undertaken at the start of each crediting period

Table 17: Regulatory framework for thermal energy supply in India

Applicable regulation	Provisions of the law	Project's compliance
Indian Standard for Portable Solid Bio-Mass Cookstove (Chulha) [First revision of 13152(Part 1):1991] ²⁰	The standard provides that the thermal efficiency for each Cookstove when tested in accordance with the method described in Annex A shall not be less than 25 percent for all types of designs and sizes of solid bio-mass natural	The stove technologies promoted by the project activities have been tested and the thermal efficiency achieved for the stoves is 28.9% for Prime stoves, 32.098% for Smart stoves and 38% for Jumbo stoves. This is above the required

²⁰ [Final Modified BIS \(Cookstove\) April 2013 MED 04\(1157\)F.pdf \(newdawnengineering.com\)](#)

	draft Cookstoves. For forced draft types it shall not be less than 35 percent. The Power Output Rating shall be between 0.5 to 3.0 kWth.	minimum of 25% provided in the Standard. The power output for smart and Jumbo stoves was tested and confirmed to be 1.34 KW and 2.32 KW respectively, which is within the allowed range. The thermal output prime stoves shall be tested and confirmed prior to first verification but is expected to be within the set range. The project has therefore complied with the requirements set out.
The National Clean Air Programme (NCAP), 2019 ²¹	This is a recent governmental initiative, launched in January 2019 by the Ministry of Environment, Forest and Climate Change (MoEFCC). The Programme focuses on the most polluted Indian cities which are to prepare action plans with time-bound targets to prevent, control, and reduce emissions.	The project activity promotes use of improved biomass stoves which will significantly reduce indoor air pollution among beneficiary households. This will commensurate the government efforts of reducing emissions from cooking and avert related household air pollution deaths and illnesses.

²¹ https://moef.gov.in/wp-content/uploads/2019/05/NCAP_Report.pdf

Data/parameter ID	SDG 13 ICS 8
Data/parameter	EF_{b,f,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 112 tCO₂/TJ Charcoal: Methodology default, 112 tCO₂/TJ (combustion only) Methodology default, 165.22 tCO₂/TJ (includes charcoal production emissions) Methodology cap, 197.15 tCO₂/TJ (includes charcoal production emissions) Other fuels: IPCC defaults When emissions from fuel production, transport, and similar are included to determine a project-specific emission factor, then the following shall apply as well: - The project boundary must include these processes - Avoidance of double counting considerations must cover all steps in the project boundary. - The determination of the specific emissions from these sources is fully documented and evidenced in the PDD These provisions may be applied to include the actual GHG emissions happening upstream in charcoal production in the charcoal emission factor; however, emission factors higher than the methodology cap are not permitted.
Value(s) applied	112
Choice of data or Measurement methods and procedures	The methodology default value of 112 tCO ₂ /TJ for firewood is applied.
Purpose of data	Calculation of baseline emissions
Additional comment	Charcoal should apply the emission factor for charcoal fuel.

Data/parameter ID	SDG 13 ICS 9
Data/parameter	$EF_{b,f,nonCO2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default: - 9.46 tCO₂e/TJ (AR5 GWP) or 8.692 tCO₂e/TJ (AR4 GWP) Charcoal: Methodology defaults: - 5.865 tCO₂e/TJ (AR5 GWP) (combustion only) - 44.83 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions), - Methodology cap: 92.29 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions) – or - 5.298 tCO₂e/TJ (AR4 GWP) (combustion only) - 40.26 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) - Methodology cap: 82.90 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) Other fuels: Any of the following, in order of preference: - IPCC defaults - Project-specific field tests prior to first verification by a qualified entity that is certified or accredited by National Standards body - Project-relevant measurement reports by qualified entities - National defaults - Credible published literature for the project area If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence. When emissions from fuel production, transport, and similar are included to determine a project-specific emission factor, then the following shall apply as well: - The project boundary must include these processes - Avoidance of double counting considerations must cover all steps in the project boundary
Value(s) applied	8.692 (AR4 GWP)
Choice of data or Measurement methods and procedures	The methodology AR4 GWP default value for firewood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter ID	SDG 13 ICS 10
Data/parameter	EF _{p,f,CO2}
Unit	tCO ₂ /TJ
Description	CO2 emission factor arising from use of fuels in project scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 112 tCO₂/TJ Charcoal: Methodology default, 112 tCO₂/TJ (combustion only) Methodology default, 165.22 tCO₂/TJ (includes charcoal production emissions) Methodology cap, 197.15 tCO₂/TJ (includes charcoal production emissions) Other fuels: - IPCC defaults
Value(s) applied	112
Choice of data or Measurement methods and procedures	The methodology default value for firewood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter ID	SDG 13 ICS 11
Data/parameter	EF_{p,f, nonCO2}
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in project scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default: - 9.46 tCO₂e/TJ (AR5 GWP) or - 8.692 tCO₂e/TJ (AR4 GWP) Charcoal: Methodology defaults: - 5.865 tCO₂e/TJ (AR5 GWP) (combustion only) - 44.83 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions), - Methodology cap: 92.29 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions) - or - 5.298 tCO₂e/TJ (AR4 GWP) (combustion only) - 40.26 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) - Methodology cap: 82.90 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) Other fuels: Any of the following, in order of preference: - IPCC defaults, - Project-specific field tests prior to first verification by a qualified entity that is certified or accredited by National Standard body - Project-relevant measurement reports by qualified entities, - National defaults - Credible published literature for the project area If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence.
Value(s) applied	8.692 (AR4 GWP)
Choice of data or Measurement methods and procedures	The methodology default value for firewood is applied
Purpose of data	Calculation of baseline emission reductions.
Additional comment	-

Data/parameter ID	SDG 13
	ICS 12
Data/parameter	$NCV_{b,fuel}$
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 0.0156 TJ/ton Charcoal: Methodology default, 0.0295 TJ/ton Other fuels: - IPCC defaults - Project-specific testing by a qualified entity that is certified or accredited - Project-relevant measurement reports by a qualified entity If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence. IPCC defaults, project-relevant measurement reports, or project-specific testing
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	The methodology default value for firewood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	The methodology default emission factor must be applied when the methodology default NCV is applied.

Data/parameter ID	SDG 13 ICS 13
Data/parameter	NCV_{p,fuel}
Unit	TJ/ton
Description	Net calorific value of the fuels used in the project
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 0.0156 TJ/ton Charcoal: Methodology default, 0.0295 TJ/ton Other fuels: - IPCC defaults - Project-specific testing by a qualified entity that is certified or accredited - Project-relevant measurement reports by a qualified entity If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence.
Purpose of data	Calculation of baseline emissions
Additional comment	The methodology default emission factor has been applied since methodology default NCV is applied.

Data/parameter ID	SDG 13 ICS 17
Data/parameter	$f_{NRB,I,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	Determined by following the CDM TOOL30 ²² , Calculation of the fraction of non-renewable biomass
Value(s) applied	0.8013
Choice of data or Measurement methods and procedures	Calculated
Monitoring frequency	One of two options, with the option defined and fixed at project design certification stage: 1. Determined ex-ante and fixed for a given crediting period or 2. Updated biennially or at each monitoring and verification PP has chosen option 1 and the parameter has been fixed ex-ante for the crediting period
QA/QC procedures	Requirements of the CDM TOOL30
Purpose of data	Calculation of baseline emissions
Additional comment	Updated values from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 4.9 ²³ have been applied. Project developers applying for a renewal of the crediting period must reassess the NRB based on most recent information available.

²² [MP83 EA08 TOOL30 \(unfccc.int\)](#)

²³ (https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf)

Data/parameter ID	SDG13
	ICS 28
Data/parameter	LE _{p,y}
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Methodology default value
Value(s) applied	0.95
Choice of data or Measurement methods and procedures	Methodology default value is applied
Monitoring frequency	Every two years, or default discount value of 0.95 applied to emission reductions
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for QA/QC
Purpose of data	Calculation of baseline emissions
Additional comment	The methodology default value of 0.95 is applied to account for any leakage and hence no monitoring will be required for the parameter. The parameter value is fixed ex-ante for the crediting period.

B.6.3. Ex ante estimation of SDG Impact

Ex ante calculations related to the outcomes of SDG 13

Emission reductions

The transparent ex-ante calculations of the outcomes of SDG 13 (i.e. CO₂e reductions) are provided in a separate Excel Spreadsheet uploaded to SustainCert App. For data/parameters available before design certification values contained in section B.6.2 above have been applied and for data/parameters not available before design certification the estimates contained in section B.7.1 below have been used.

The estimated emission reduction for the project activity are as follows:

Jumbo stoves					
Year	Stove Numbers	Baseline estimate	Project estimate	Leakage emissions	Net benefit
29/10/2020 to 31/12/2020	116	86	35	0.04	51
2021	15,126	67,194	27,408	0.22	39,786
2022	33,383	148,297	60,490	0.22	87,807
2023	33,383	148,297	60,490	0.22	87,807
2024	33,383	148,297	60,490	0.22	87,807
01/01/2025 to 28/10/2025	33,383	123,581	50,408	0.19	73,173
Totals		635,752	259,320	1.11	376,431
Average		127,150	51,864	0.222	75,286

Source: GS 11358 ER spreadsheet tab project ERs

Prime stoves					
Year	Stove Numbers	Baseline estimate	Project estimate	Leakage emissions	Net benefit
29/10/2020 to 31/12/2020	0	0	0	0.04	0
2021	4,383	19,471	10,127	0.22	9,343
2022	6,260	27,809	14,464	0.22	13,344
2023	6,260	27,809	14,464	0.22	13,344
2024	6,260	27,809	14,464	0.22	13,344
01/01/2025 to 28/10/2025	6,260	23,174	12,054	0.19	11,120
Totals		126,071	65,574	1.11	60,496
Average		25,214	13,115	0.222	12,099

Source: GS 11358 ER spreadsheet tab project ERs

Smart stoves					
Year	Stove Numbers	Baseline estimate	Project estimate	Leakage emissions	Net benefit
29/10/2020 to 31/12/2020	261	193	93	0.04	100
2021	1,355	6,019	2,907	0.22	3,113
2022	2,086	9,267	4,475	0.22	4,792
2023	2,086	9,267	4,475	0.22	4,792
2024	2,086	9,267	4,475	0.22	4,792
01/01/2025 to 28/10/2025	2,086	7,722	3,729	0.19	3,993
Totals		41,735	20,152	1.11	21,581
Average		8,347	4,030	0.222	4,316

Source: GS 11358 ER spreadsheet tab project ERs

Combined totals					
Year	Stove Numbers	Baseline estimate	Project estimate	Leakage emissions	Net benefit
29/10/2020 to 31/12/2020	377	279	128	0.11	151
2021	20,864	92,684	40,442	0.67	52,241
2022	41,729	185,372	79,428	0.67	105,943
2023	41,729	185,372	79,428	0.67	105,943
2024	41,729	185,372	79,428	0.67	105,943
01/01/2025 to 28/10/2025	41,729	154,477	66,190	0.56	88,286
Totals		803,557	345,046	3.33	458,508
Average		160,711	69,009	0.666	91,702

Source: GS 11358 ER spreadsheet tab project ERs

Ex ante calculations related to the outcomes of SDG 1

Estimated monetary savings computed from fuel saved (Rupee/household/day)

The VPA includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cooking methods and thus result in a reduction in firewood consumption. Using the project technology is expected to lead to financial savings through reduced fuel consumption costs.

The ex-ante estimate for SDG 1 has been calculated based on estimated household savings by assessing the prevailing fuel prices in the host country. This was calculated as follows:

Monetary savings	Value
Average daily expenditure on fuel (baseline)	37.41
Expenditure with project stove	15.90
Savings (Rupee)	21.51

Source: GS 11358 ER spreadsheet tab specific savings

The project is expected to lead to saving of about 21.51 Rupee/household/day on fuel costs.

Ex ante calculations related to the outcomes of SDG 5

Average time saved by women for fuel collection (hours/household/day)

The use of a Greenway stove in a household helps in reducing the time spent on firewood collection which is mostly done by women. Fuel collection takes more time and effort, which takes away productive hours of women and young girls leading to time poverty and lost opportunity. In the baseline, 55.5% of the households spent between 4-8 hours per day in fuel collection. The ex-ante estimation for the parameter has been determined by establishing the average hour $(4+8/2)$ to come up with a value of 6 hours per day.

By using the project technology, the number of hours spent per household in fuel collection is expected to be about 2.55 which has been calculated by factoring in the efficiency of both the baseline and project stoves.

The estimated time saving is calculated as the difference between time spent in the baseline less time spent in the project scenario fuel collection. The time savings has been calculated to be 3.45 hours/household/day. The ex-ante estimation for the parameter has been determined as follows:

Time saving	Value
Average daily time spent in baseline fuel search	6.00
Average daily time spent in project fuel search	2.55
Savings hrs)	3.45

Source: GS 11358 ER spreadsheet tab specific savings

Ex ante calculations related to the outcomes of SDG 7

Number of people/households with access to basic service provided by the project device

This ex-ante estimation for the parameter has been derived from the projected stove sales provided by the project. The project is expected to distribute 41,729 improved fuelwood cookstoves in total among household end-users as summarized below.

Year	Jumbo Stoves	Prime Stoves	Smart stoves	Total
29/10/2020 to 31/12/2020	116	0	261	377
2021	15,126	4,383	1,355	20,864
2022	33,383	6,260	2,086	41,729
2023	33,383	6,260	2,086	41,729
2024	33,383	6,260	2,086	41,729
01/01/2025 to 28/10/2025	33,383	6,260	2,086	41,729

Source: GS 11358 ER spreadsheet tab project ERs

Ex ante calculations related to the outcomes of SDG 8

Total number of jobs split by gender

The VPA is expected to job creation through the stove production and distribution process. This ex-ante estimation for this parameter has been determined from the estimate provided by the project implementer. Each year the project is expected to employ about 300 people including 200 males, and 100 females.

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13

Year	Baseline estimate (tCO ₂ e)	Project estimate (tCO ₂ e)	Leakage estimate (tCO ₂ e)	Net benefit (tCO ₂ e)
29/10/2020 to 31/12/2020	279	128	0.11	151
2021	92,684	40,442	0.67	52,241
2022	185,372	79,428	0.67	105,943
2023	185,372	79,428	0.67	105,943
2024	185,372	79,428	0.67	105,943
01/01/2025 to 28/10/2025	154,477	66,190	0.56	88,286
Total	803,557	345,046	3.33	458,508
Total number of crediting years		5		
Annual average over the crediting period	160,711	69,009	0.666	91,702

SDG 1: Estimated monetary savings computed from fuel saved
(Rupee/household/day)

Year	Baseline estimate	Project estimate (rupee/hh/day)	Net benefit (rupee/hh/day)
29/10/2020 to 31/12/2020	6.23	2.6	3.6
2021	37.41	15.9	21.5
2022	37.41	15.9	21.5
2023	37.41	15.9	21.5
2024	37.41	15.9	21.5
01/01/2025 to 28/10/2025	31.17	13.2	17.9
Total	187.03	79.5	107.5
Total number of crediting years	5		
Annual average over the crediting period	37.41	15.9	21.5

Source: GS11358 SDG Impact tool tab Greenway SDG Impact assessment

SDG 5: Average time saved by women for fuel collection
(hours/household/day)

Year	Baseline estimate	Project estimate (rupee/year/household)	Net benefit (rupee/year/household)
29/10/2020 to 31/12/2020	3.0	1.28	1.725
2021	6.0	2.55	3.45
2022	6.0	2.55	3.45
2023	6.0	2.55	3.45
2024	6.0	2.55	3.45
01/01/2025 to 28/10/2025	3.0	1.28	1.725
Total	30.0	12.75	17.25
Total number of crediting years	5		
Annual average over the crediting period	6.0	2.55	3.45

Source: GS11358 SDG Impact tool tab Greenway SDG Impact assessment

SDG 7: Number of households with access to basic service provided by the project device

Year	Baseline estimate	Project estimate (number)	Net benefit (number)
29/10/2020 to 31/12/2020	0	21,241	21,241
2021	0	41,729	41,729
2022	0	41,729	41,729
2023	0	41,729	41,729
2024	0	41,729	41,729
01/01/2025 to 28/10/2025	0	41,729	41,729
Total	0	41,729	41,729
Total number of crediting years	5		
Annual average over the crediting period	0	41,729	41,729

Source: GS11358 SDG Impact tool tab Greenway SDG Impact assessment

SDG 8: Total number of jobs split by gender

Year	Baseline estimate	Project estimate (number)	Net benefit (number)
29/10/2020 to 31/12/2020	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
2021	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
2022	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
2023	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
2024	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
01/01/2025 to 28/10/2025	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
Total	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)
Total number of crediting years	5		
Annual average over the crediting period	0	300 people (200 males, 100 females)	300 people (200 males, 100 females)

Source: GS11358 SDG Impact tool tab Greenway SDG Impact assessment

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter ID	SDG 13
	ICS 15
Data / Parameter	Avoidance of double counting or double claiming among project technology end users
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with project technology end-users
Source of data	Carbon title waiver forms signed by end-users transferring their rights to Greenway Grameen Infra Private Limited
Value(s) applied	No double counting. Evidence of carbon waiver transfer to be provided during verification.
Measurement methods and procedures	NA
Monitoring frequency	Monitored whenever project technology is sold
QA/QC procedures	Cross check using general internet search and search of public records of Gold Standard and other voluntary market and UNFCCC mechanisms
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter ID	SDG 13 ICS 16
Data / Parameter	Presence of stove stacking
Unit	NA
Description	Descriptive statistics of the presence and usage practices of baseline- and other non-project-technology by project technology end users.
Source of data	The methodology provides that any one of the following methods may be applied: - Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity, or - Usage Survey- use of other stoves, to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices, or monitoring surveys to capture the number of meals cooked. The surveys may be integrated with the usage survey The PP will use usage Survey to assess this parameter
Value(s) applied	0
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	The calculation of $SFS_{p,b,y}$ shall be cross-checked with the observed presence of stove stacking. Ensure any stove stacking is considered so that emission reductions are calculated only from real reduction of, or replacement of, baseline fuel use.
Purpose of data	Calculation of emission reductions
Additional comment	As required under the methodology, whether or not the existing baseline technology is surrendered, when an old technology remains in use in parallel with the improved technology, or another technology is put in use in parallel, the corresponding emissions shall be accounted for so that emission reductions are not overestimated. The PP assumes that there will be no presence of stove stacking in the project scenario and a value of 0 is applied for ex-ante estimation purposes. Further, each household is only entitled to one project stove type to ensure there is no double counting. The number of stoves sold to each household shall be recorded on the sales database and cross-checked during monitoring.

Data / Parameter ID	SDG 13 ICS 18
Data / Parameter	P _{b,y}
Unit	kg/household-day
Description	Quantity of fuel that is consumed in baseline scenario b during year y
Source of data	Baseline survey ²⁴
Value(s) applied	7.93
Measurement methods and procedures	Measured
Monitoring frequency	To be updated before the next verification and thereafter be fixed for one crediting period.
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4), general requirements for QA/QC (Section 4.5) and Annex 2 Kitchen performance test.
Purpose of data	Calculation of emission reductions
Additional comment	Used to calculate SFS under method 1 Applicable adjustment factors may be applied. A kitchen performance test (KPT) shall be conducted prior to the first verification and the value achieved will be updated and fixed for the crediting period.

²⁴ The value is obtained from the baseline survey assessment and is applied for purposes of ex-ante estimate. A KPT shall be done prior to first verification to determine the value to be applied during the crediting period.

Data / Parameter ID	SDG 13 ICS 19
Data / Parameter	$P_{p,y}$
Unit	kg/household-day
Description	Quantity of fuel that is consumed in project scenario p during year y
Source of data	Project performance field tests
Value(s) applied	3.23 (Jumbo) 3.83 (smart) 4.12 (prime)
Measurement methods and procedures	Measured
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4), general requirements for QA/QC (Section 4.5) and Annex 2 Kitchen performance test.
Purpose of data	Calculation of emission reductions
Additional comment	Used to calculate SFS under method 1. Applicable adjustment factors may be applied. The value applied is an estimate, actual value to be determined during monitoring.

Data / Parameter ID	SDG 13 ICS 20
Data / Parameter	$SFS_{b,p,y}$
Unit	tonnes/unit/year
Description	Specific fuel savings for an individual project technology in baseline b/project p pair in year y
Source of data	Calculated from $P_{b,y}$, $P_{p,y}$ and other information to obtain the savings in the required units
Value(s) applied	1.71 (Jumbo) 1.5 (smart) 1.39 (prime)
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method, inputs and their sources shall be described in detail in the VPA-DD and monitoring report. Cross-check with proportional efficiency of baseline and project technology. For example, if the baseline stove efficiency is 10% and the project rated efficiency is 25%, the savings should reflect a factor of 2.5 approximately in the first year of the crediting year and not more. If it is more, then the value is capped based on the proportional efficiency.
Purpose of data	Calculation of emission reductions
Additional comment	Applies when using Method 1. The value applied is an estimate, actual value to be determined during monitoring.

Data / Parameter ID	SDG 13 ICS 26
Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Weighted average usage rate in project scenario p during year y
Source of data	Usage survey exercise
Value(s) applied	100
Measurement methods and procedures	Calculated. The survey result shall provide the statistically valid proportion of users actively using the project technology for each project technology age cohort. From the annual usage survey results, the weighted average percent of users actively using the project technology will be calculated, where the weighting is by the quantity of project technologies of each age cohort being credited in a given project scenario.
Monitoring frequency	At least annually Optionally, more frequently
QA/QC procedures	Compliance with the general requirements for sampling (Section 4.4) and general requirements for QA/QC (Section 4.5)
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante emission calculations. Actual value to be determined during monitoring. Please refer to the Requirements and Guidelines: Usage Rate Monitoring for carrying out usage surveys for projects implementing improved cooking devices.

TEMPLATE

Data / Parameter ID	SDG 13 ICS 27
Data / Parameter	$N_{b,p,y}$
Unit	days
Description	Number of project technology-days included in the project database for baseline b/project p pair in year y
Source of data	Calculated from the Project database as the sum of the number of project technology units times the calendar days during the year that they were present at the end user locations
Value(s) applied	365
Measurement methods and procedures	Calculated
Monitoring frequency	Annually
QA/QC procedures	Cross check the results of the usage survey with the contents of the project database to confirm whether the project technology units surveyed are present at end user locations as expected, or not. If there is discrepancy, this must be explained or corrected.
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

SDG 1

Data / Parameter	SDG 1 No poverty
Unit	Rupee/household/day
Description	Number of people/households with access to basic service provided by the project device
Source of data	Project surveys
Value(s) applied	21.51
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the proportion of population who demonstrate achieving monetary saving from use of the project technology.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

SDG 5

Data / Parameter	SDG 5 Gender Equality
Unit	hours/household/day
Description	Average time saved by women from fuel collection.
Source of data	Project records and surveys
Value(s) applied	3.45
Measurement methods and procedures	Surveys
Monitoring frequency	Annual
QA/QC procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the average time saving associated with fuel collection among end-users.
Purpose of data	Reporting on SDG impacts
Additional comment	The baseline survey finding also indicate that about 55.5% of the respondent reported spending 4-8 hours in fuel collection. Using the project technology is expected to reduce this time by about 3.45hours. Value applied is for purposes of ex-ante estimation. Actual value to be determined during monitoring.

SDG 7

Data / Parameter	SDG 7 Access to clean energy
Unit	Number
Description	Number of people/households with access to basic service provided by the project device
Source of data	Project sales database
Value(s) applied	Jumbo: 33,383 Smart: 2,086 Prime: 6,260 Total: 41,729
Measurement methods and procedures	The sales records will be checked to determine the number of people using the project technology.
Monitoring frequency	Annual
QA/QC procedures	An updated sales database shall be maintained by the project implementor.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is the projected number for the first crediting period. Actual number will be determined during monitoring.

SDG 8

Data / Parameter	SDG 8 Decent Work and economic growth
Unit	Number
Description	Total number of jobs split by gender
Source of data	Employment records
Value(s) applied	300 people (200 male, 100 female)
Measurement methods and procedures	.To be determined from employment records
Monitoring frequency	Annual
QA/QC procedures	The project implementor shall maintain up to date employment records to confirm the number of people employed per department, split by gender.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is the projected number. Actual number will be determined during monitoring.

B.7.2. Sampling plan

Sampling Objective and Parameter to be monitored

Section 4.4. of the applicable methodology outlines the general requirements for sampling and provides that when sampling is applied to determine mean (average) parameter values or proportion (yes/no) parameter values for both ex-ante and monitored data and parameters, the outlined guidelines shall always be applied. Furthermore, specific requirements apply for the sampling related to some parameters.

The sampling method chosen for the project activity is Stratified Random Sampling. This will be done in accordance with the Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4. The sampling criteria has been found to be suitable since the project activity shall distribute different models of the project technologies.

The project developer shall maintain an updated distribution record for all the units, and this will identify the end-users and the geographical zones of operational project

technologies within the project activity. Random test subjects shall hence be identified from the distribution.

The parameters to be monitored by the VPA will be the fuel consumption trends, drop off rates and project usage rates. The target population for the proportion of stoves in operation ($U_{p,y}$), presence of stove stacking and fuel consumption and savings ($P_{p,b,y}$) for this VPA is all the households in the VPA database which are using Greenway stoves being distributed. The determination of the presence of stove stacking will be conducted together with the usage survey.

Target population

The target population will be the total stove population in the VPA.

Sampling Methodology

One unit of baseline cookstove will be replaced in each household with one type of Greenway stove in this project. Since the cooking patterns and behaviours are the same across the VPA boundary, the usage pattern will most likely be similar. The VPA sells three different stove sizes and due to differences in stove sizes, the project will employ a stratified random sampling method across the VPA when drawing up a sample. When picking individual samples from each stratum, simple random sampling will be applied.

Sample Size

The minimum sample sizes for the determination of the monitored parameters will be determined as per 'The Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities', Version 04.0 and the registered PoA-DD. The minimum sample sizes required for the determination of the different types of parameters (proportion and mean) will be determined using different methods as described below.

U_{p,y} survey sample size calculation (proportion parameter), presence of stove stacking and Specific fuel savings for an individual technology of project (P_{p,b,y}) (mean parameter)

Sampling can be done per VPA or cross-VPA sampling if more than one VPA of similar nature are implemented and for cross-VPA sampling the required applicable confidence level of 95/10.

When determining the values of interest i.e. proportion parameter and mean parameter, the following equation for the sample size will be used to calculate sample size when applying stratified random sampling (*As provided in the Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities', Version 04.0*).

Proportion parameter

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 p(1-p)} \quad \text{Equation (1)}$$

Mean parameter.

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V} \quad \text{Equation (18)}$$

Baseline and project survey

When a baseline and project survey is used the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller.
- Group size 300 to 1000: Minimum sample size 10% of group size
- Group size > 1000 Minimum sample size 100

General Requirements for Quality Assurance and Quality Control

- i. The project developer is responsible for accurate and transparent record keeping, monitoring and evaluation. All supporting documentation and records for the project must be easily accessible for spot checking and cross referencing by a third-party auditor.
- ii. Contact information in the total sales/distribution record shall allow a project auditor to easily contact and visit end users.
- iii. An auditor must be able to cross reference pertinent project documentation, including archives such as production records (e.g., materials purchases, internal logs), financial accounts and sales records, as well as wholesale customer invoices, observations of retailer activities and sales performance.

B.7.3. Other elements of monitoring plan

The following shall be monitored and recorded during project crediting period.

a) Total sales or dissemination record

The project developer shall maintain an accurate and complete "dissemination record" or "installation record." The record shall be maintained continuously and backed up electronically. The required data for each project unit is:

1. Date of installation
2. Geographic area of sale
3. Model/type of project technology sold
4. Quantity of project technologies sold
5. Name, telephone number (if available), and address and/or GPS coordinates: a. Required for all bulk purchasers, i.e., retailers and industrial users; b. Required for all end users, e.g. households
6. Mode of use: domestic, institutional commercial, other.

b) Project Database

The project database lists all the project technology units that have been distributed by the project and have not surpassed their technical life. It will be derived from the total dissemination record and shall be maintained continuously. Within the database, project technologies units are labelled, at a minimum, with their corresponding project

scenario p and their date of dissemination. Technologies aged beyond their technical life, and not replaced or retrofitted, are removed from the project database, and no longer credited.

c) Project Surveys

The project surveys include the usage survey ($U_{p,y}$), Usage Survey- use of other stoves (ICS 16), and the baseline and project performance field tests (BFT and PFT).. The project surveys have the same sample sizing and data collection guidelines as the baseline scenario survey. The project surveys shall only be conducted with end user's representative of the project scenario and currently using the project technology.

i) Usage Survey ($U_{p,y}$)

The objective of the usage survey is to provide a single usage parameter that is weighted based on the age distribution for project technologies in the Project Database, $U_{p,y}$. This usage parameter must be established to account for drop off rates as project technologies age and are replaced, or for reduced use by end users for other motives.

The usage survey shall determine the usage proportion for each project technology being credited for each project scenario p. The usage survey shall be conducted in line with the requirements and guidelines of sampling.

ii) Usage Survey- use of other stoves

As part of the usage survey, the project developer shall collect data on the presence and usage practices of baseline and other non-project technology by project technology end users and prepare descriptive statistics of these practices. The same method of in person interviews and expert observation within the kitchen in question will be used to collect these data. Where the usage survey indicates usage of other non-project technology, this shall be addressed in line with the requirements for parameter ICS 20.

iii) Field Tests

The baseline and project performance field tests (BFT and PFT) measure real, observed technology performance in the field. Since the project applies method 1, a field test shall be carried out both for baseline and project scenarios by testing an independent sample (different subjects for baseline and project scenarios).

The approach taken to conduct the performance tests will ensure:

- It is transparent and can easily be replicated.
- It is evidently conservative.
- The sample is randomly selected to avoid material bias.
- The impact of daily and seasonal variations on the expected average fuel consumption savings is accounted for.

The project developer shall design the field test to ensure monitoring is representative of technology and fuel use practices. The end-users will be required to behave and consume fuel normally, using the technologies they normally use. Participants will not be influenced to use a specific stove or fuel during the study, nor deviate from typical stove and fuel practices.

d) Baseline Fuel Consumption

To calculate SFS under method 1 applied, the project developer shall determine $P_{b,y}$, Quantity of fuel that is consumed in baseline scenario b during year y in units of kg/household-day, kg/person-meal, etc. The baseline fuel consumption shall be determined using project specific Baseline Performance Field Tests (BFT). The BFT will be completed once ex-ante, and at crediting period renewal. The following requirements shall apply for selecting the end user locations of the Baseline Performance Field Tests:

- The BFT shall reflect the average household size within the project area.
- The BFT shall take place in those households where improved devices have not yet been disseminated and where baseline technology is still in use in the project area.
- Where all households in the project area already use exclusively improved devices, the BFT is performed in households in the vicinity of the project target area that exhibit the same socio-economic circumstances as the households that have already received the improved devices in the project target area and

are operating devices that are similar to the baseline technology in the project activity.

e) Project Fuel Consumption

To calculate SFS under method 1 applied, the project developer shall determine $P_{p,y}$, Quantity of fuel that is consumed in project scenario p during year y in units of kg/household-day, kg/person-meal, etc. The project fuel consumption shall be determined using a project specific Project Performance Field Test (PFT). The PFT is completed every other year, or more frequently, and the first PFT must be completed before the first verification of the project. The following requirements shall apply for selecting the end user locations of the Project Performance Field Tests:

- End users shall be representative of the project scenario target population and currently using the project technology.
- The PFT is a study of the fuel consumption of the cooking practices of project technology end users during the crediting period. The PFT shall evaluate whether the project technology provides the same services at the same/similar frequency as the baseline technology. As such, when the project technology does not replace or reduce the entire baseline fuel consumption, the calculation of SFS shall adjust the baseline fuel consumption downward based on the cooking services replaced by the project stove.
- Where the baseline technology or other non-project technologies operate as backups or complementary units in parallel with project technology ("stove stacking"), the project fuel consumption implications shall be accounted for in the PFT.

The following will be determined through sampling in accordance with the "Standard for sampling and surveys for CDM project activities and Programme of Activities", Version 09.0 and the methodology TPDDTEC, ver 4.0:

- Mean fuel savings ($P_{p,b,y}$)
- The usage rate ($U_{p,y}$)

If annual inspection is to be applied, a 90% confidence interval and a 10% margin of error will be achieved for the sampling parameters above. In case the 90/10 precision is not achieved, the lower bound of a 90% confidence interval of the parameter value

will be chosen. However, where cross VPA sampling plan is applied as referred to in paragraph 22 of the Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, Version 09.0, 95/10 confidence/precision will be applied for the sample size calculation. In case the 95/10 precision is not achieved, the lower bound of a 95% confidence interval of the parameter value will be chosen.

f) Data Archiving

Data collected for the purposes of monitoring the VPA will be archived in electronic format and where possible in hard copies. The same will be availed to VVB during verification when required.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

29/10/2020 (date when the first unit of the project stove was sold)

C.1.2. Expected operational lifetime of project

15 years 0 Months

C.2. Crediting period of project

C.2.1. Start date of crediting period

29/10/2020 ²⁵(start of the project)

C.2.2. Total length of crediting period

5 years 0 Months (renewable twice)

²⁵ As per the GHG requirements para 10.2.1, the start date of Crediting Period is the date of start of operation or a maximum of two years prior to the date of Project Design Certification, whichever occurs later.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 2	No mitigation measures proposed as the project is designed to enhance gender equality. During monitoring, the number of people employed by the project will be monitored and be desegregated by occupation and gender.
Principle 3	The PD shall ensure workers are trained to minimise on potential risks. Training records to be provided as supporting evidence.
Principle 7	No mitigation measures proposed. During monitoring, the project will check the usage rates and the operating efficiencies of the stove to determine the emission reductions achieved by the project.
Principle 6	No mitigation measures proposed but the project will monitor the adherence to labour rights and ensure that all workers are treated fairly and equally as per the company's non-discrimination policy.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project has a policy which requires non-discrimination of employees and put measures in place for equal treatment to all its workers. The non-discrimination covered by the policy encompasses discrimination based on gender. This policy will continue to be enforced and appropriate measures will be undertaken to ensure that the same is complied with. The policy provides for means of addressing discrimination whenever it arises.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project developer abides by the principle of gender equality as enshrined in the Indian Constitution in its Preamble, Fundamental Rights, Fundamental Duties and Directive Principles. The Constitution grants equality to women. The project will, therefore, aim to abide by the constitution and any applicable laws. The non-discrimination policy of equal opportunity to all aligns well with constitution requirements.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

E.1. Summary of stakeholder mitigation measures

The project carried two rounds of consultations; the first round of consultation entailed physical meetings while the second round of consultation was the online stakeholder feedback round (please refer to the separate Stakeholder Consultation Report for a detailed report). The project conducted four physical stakeholders' consultation meetings in different places targeting all stakeholder categories in line with the GS4GG stakeholder consultation and engagement requirements. The first stakeholder meeting was held at Annex GIDC Savli, Vadodara Rural, Gujarat on 14th October 2020, while the second one was held on 11/11/2020 at Janpad Office, Block, Lovkushnagar, District, Chattarpur. M.P. The third meeting was held on 29th January 2021 at Savli, Vadodara Rural, Gujarat 391775. and the last meeting was held on 5th March 2021 at Kanfola Bari, Block: Burbandha, District: Morigaon, Assam. The target stakeholders were end-user, retailers and financial institutions, local community, representatives of local NGOs and government agencies and the local authority. The consultations were held during the Covid pandemic hence they were organised in smaller groups due to COVID-19 restrictions in the host country.

From the consultations held, the following is the summary of the comments and mitigation measures proposed:

- A member of a local women's self-help group expressed that women greatly suffer due to cooking on traditional stoves as they expend a majority of their time in collection of firewood and cooking which restricts them from engaging in any other productive activities that can add to the livelihood generation of their respective households. They also get tired and incur negative health impacts which adds to their burden. Hence adopting a clean cooking stove which can reduce firewood consumption and household smoke emission will be beneficial to rural women.
- What is the lifespan of the stove and how to deal with any breakage of the parts?

Answer: Greenway stoves have a demonstrated lifespan of about 6 years, and depending on the maintenance, can also last for up to 7 years. The stoves or any parts thereof will be replaced if there is a breakage. However, some replacements can only be done at a cost if the warranty period for the stove is over.

- Can the stove be cleaned for spillage?

Answer: The stove can be wiped clean with a wet cloth after being used and after the body has cooled down.

- How much weight can the stoves bear?

Answer: The stoves have a pot loading capacity of 25kg to 40 kg.

- Who calculates and checks the value claimed for GHG reductions on stoves?

Answer: The calculation is done as per methodologies prescribed by the Gold Standard, these have some assumed values and some other variables for which measurements need to be done such as stove efficiency. This methodology is publicly available for anyone to read. The checking is done by a third-party auditor.

- Is the checking done on all stoves? Or a subset?

Answer: The checking is done on a drawn sample of the users of the stoves.

- Is there any government support or subsidy also made available under this program?

Answer: No, there is no government support or subsidy available on the stoves or related activities

- What is the tax rate on stoves? Is there any concession available on GST?

Answer: No. A standard rate of 12% GST is applicable on efficient stoves.

- How far an improvised firewood is stove relevant despite provision of LPG connection to households under Ujjwala Scheme?

Answer: Even though rural households have been provided with LPG connection

under Ujjwala scheme, the complete adoption of LPG is still a farfetched goal as many communities still use polluting tradition stoves for a good majority of their cooking and boiling/heating water due to high cost of refilling LPG cylinders. Also, they may have to travel to some distance away from their village to reach a distribution centre to get their LPG cylinders refilled which may anywhere between a few hours to a few days and in the meanwhile they may have use to firewood for cooking. Greenway Stove acts as an intermediary level in the transition from traditional stove to LPG for most households who have adopted Greenway Stove

- Does adopting an improved firewood stove such as Greenway stove promote, felling of trees since it also uses firewood as fuel?

Answer: Greenway or its product is in no way promoting felling of trees. Greenway stoves have improved efficiency which ultimately reduce fuel consumption to a level where the firewood needs of a household are greatly reduced. The project specifically aims to replace usage of traditional stoves and those types of stoves only with the Greenway Stove thereby reducing fuel consumption from the mud stove as a baseline. Greenway Stoves are not intended replacements for other fuel types such as electricity, solar, LPG etc

- How does Greenway plan to promote the adoption of Clean Cooking through high efficiency cook stoves as is the objective of the said project, amongst economically weaker households?

Answer: To make Greenway Stove available to and affordable by everyone, the project provides for Carbon Financing through which customers are offered upfront subsidy of flexible repayment on the stove. Along with this the customers are also offered the stove in easy-to-pay EMIs of 3-6 months.

- A user of traditional stove commended the performance of the stove as she had used such a stove for the first time during the demonstration. She reaffirmed by her own experience that the stove was effective for smoke reduction and the fuel used and the time taken to cook was significantly reduced.

- A member of a local self- help group highlighted that this stove can be especially useful in Morigaon as the area is flood prone and given the portability of the stove, it can come handy during times of emergency.
- A representative of RGVN, a local NGO voiced his concerns regarding the price of the product as it was too high for most rural families in the region and said that even if Greenway Stove may prove to be beneficial to the families, they may not have the paying capacity to adopt it. Hence in order to promotion the adoption such technology, it must be supplied at a substantially reduced price. To address this concern, the price of the stove is set after considering several factors such as production cost and material prices. In order to ensure that more households afford it, the stove is being provided through a loan facility whereby end users have a flexible repayment plan which enables households to pay monthly and the monthly repayment enables those with low earning capacities to afford the stove.

Stakeholder Feedback Round

The second round of stakeholder consultation was carried out in compliance with the GS requirements which require that projects shall carry out two rounds of consultations. The requirement stipulates that at least once one of the consultations should be a physical meeting while the other can be physical or be done online.

The Greenway Stakeholder Feedback Round was carried out during the COVID-19 pandemic and as such, it was not possible to carry out a physical second round of review. Instead, the project employed an online process for stakeholders who had access to email. For those without emails, the PP decided to send hard copies of the consultation report to partner organization offices and request the stakeholders to visit those offices and review the report.

However, because of the escalation of the COVID 19 pandemic in India, the process of placing a hard copy of documents for review by those without emails could have contributed to risk exposure to COVID-19 and the project did not want to contribute to or increase the exposure risks.

The SFR process was held between the period 08/07/2021 to 07/09/2021. Emails were sent out on 8th July 2021 to stakeholders who had been invited to the meeting. The documents sent out to the stakeholders included the stakeholder's consultation

report, draft PoA-DD, and draft VPA-DD. The stakeholders were requested to review the documents and provide comments before the expiry of the 60-day period. The project implementer followed up with an email reminder on 25th August 2021 requesting them to provide their feedback. However, from all the emails sent out, the project did not receive any feedback from those contacted and the project concluded that the stakeholders did not have any additional comments.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Any stove problem to be reported to nearest retail or repair centre. Stove repayment challenges to be reported to Greenway focal point/Local representative of Greenway where they make their monthly repayment.
Telephone access (optional)	+91-22-60522638 Almost all end users have access to a telephone and this method was preferred as it will enable the project team to collect feedback more rapidly.
Internet/email access (optional)	contact@greenwayappliances.com Stakeholders with internet access can use this method to raise issues/comments about the project via email.
GS Contact (mandatory)	help@goldstandard.org
Other	Alternatives, the end user can visit the Greenway Grameen Infra Private Limited office and report their grievance in person

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The constitution of the host country (India) considers it a legal offence to violate human rights during any business activity. India endorses the United Nations Guiding Principles (UNGPs) on Business and Human Rights adopted in the UN Human Rights Council (UNHRC) in 2011. The CME complies with the legal requirements of the host country. This is not violated at any point during the project.	Not Required
2. The Project shall not discriminate with regards to participation and inclusion	No		

		The parameter will not be monitored. The project targets to sell stoves to domestic users by extending credit facilities to them, they are able to afford the stoves. Any willing end user can access the stove and their will not be any form of bias when selling the stoves. The parameter will not be monitored.	Not Required
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	No	1. The project enhances women's access and entitlement of benefits. Since the women will be direct user of the stoves, it will benefit women by reducing their exposure to the indoor air pollution thereby improving their health.	Not Required
2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work	No		
	No		

3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks (where required) 4. Summary of opinions and recommendations of an Expert Stakeholder(s)	No	In addition, the decrease in quantity of firewood required will reduce workload of women for the collection of firewood. Reduced workload for firewood collection results in time saving that the women can use for other productive activities. The project will contribute the above-mentioned benefits, regardless of the marital status of women. The The parameter will not be monitored. 2. The project will not adversely affect men and	
---	-----------	---	--

		women in marginalized or vulnerable communities. Implementation of the project will contribute towards preservation of common resources in form of “firewood”. Households’ duties related to firewood collection, cooking and cleaning utensils predominantly remain with women. The project therefore tends to decrease burden on women and would not result in social isolation of men. The parameter will not be monitored. 3. The project duly accounts for the gender roles. Time saving is one of the key	
--	--	--	--

		benefits from the project which the beneficiary can utilize to fulfil their gender roles. With the saved time, one can perform the respective role more effectively. The parameter will not be monitored.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Potentially	The project activity entails manufacture and distribution of Greenway stoves among household end-users. The manufacture of the stoves is done in a modern facility and uses efficient technology to produce the stoves. The factory where the stoves are	The PD shall ensure workers are trained to minimise on potential risks. Training records to be provided as supporting evidence.

		produced has been designed to maximise on the health, safety and working conditions for the workers. The project implementer shall ensure that the workers are trained and served with the appropriate working tools and protective gear to ensure their safety and health is safeguarded.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project units will be simple and small in dimension. The project will not result in any change in cooking habits because of substitution of fuel. Therefore, the result of this project won't damage or remove cultural	Not Required
>>			

		heritage, since the project is implemented in the households of families. The parameter will not be monitored.	
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project will not involve relocation both physical or economic. The parameter will not be monitored	Not Required
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The project does not involve acquisition nor does it involve land use or tenure changes. The parameter will not be monitored	Not Required
>>			
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The project is not expected to involve any corrupt practices or reinforce the same. The parameter will not be monitored	Not Required
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	No	Greenway employees are employed, and they sign contracts outlining their roles and benefits. The contracts also outline their job descriptions. The working conditions follow laid down guidelines. All the health and safety related laws are abided by in the production and distribution process. Hence, this safeguarding principle is not triggered.	Not Required

2. Workers shall be able to establish and join labour organisations	No	The parameter will not be monitored. All workers under direct control of Greenway Appliances are subject to labours laws of India and are free to join unions of their choice.	
3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND	No	The stoves are manufactured in Greenway factory which employs only adults. The distribution of stoves is done by Greenway stoves. All the above activities does not involve use of child labour. The parameter will not be monitored.	

d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of	No No	The working conditions of the employees and workers are in compliance with national guidelines of the host country (India). Hence, this safeguarding principle will not be triggered.	
--	---------------------	--	--

appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		The parameter will not be monitored. As per occupational and safety guidelines, all workers using any equipment are trained on how to handle that equipment and any incident is documented.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project distributes stoves which help end users reduce their fuel consumption. By reducing their fuel consumption, their associated expenses on fuel collection and purchases is reduced. Therefore, the project will not cause negative economic consequences during and after implementation. In	Not Required
>>			

		addition, the project through carbon revenues has developed a favourable stove purchase and repayment plan which helps end users purchase the stove and pay in instalments. The parameter will not be monitored.	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario? >>	No	The project will deploy efficient cooking stoves against the traditional cooking stoves. Their use will ultimately result in GHG emissions reduction The parameter will be monitored. Means of verification: Annual monitoring.	Not Required

		Indicator: Continuous use of Cookstoves.	
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project involves the dissemination of units of cookstoves which does not involve energy supply. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project doesn't involve any activity related to discharge of waste-water, extraction of surface or ground water. This	Not Required

>>		safeguarding principle will not be triggered. The parameter will not be monitored.	
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project involves the dissemination of units of cookstoves which does not involve erosion or waster body instability. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project involves the dissemination of units of cookstoves which does not involve landscape	Not Required
>>			

		modification. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project involves the dissemination of units of cookstoves which does not lead to increased vulnerability of people within the project boundary. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.3 Genetic Resources			

Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The project involves the dissemination of units of cookstoves which does not involve genetic resources. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project involves the dissemination of units of cookstoves which does not involve release of pollutants. Therefore, the safeguarding principle under consideration will not be triggered by the project.	Not Required
>>			

		The parameter will not be monitored.	
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The project involves the dissemination of units of cookstoves which does not involve hazardous and non-hazardous waste. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project involves the dissemination of units of cookstoves which does not involve pesticides and fertilisers. Therefore, the safeguarding principle under	Not Required
>>			

		consideration will not be triggered by the project. The parameter will not be monitored.	
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project involves the dissemination of units of cookstoves which does not involve harvesting of forests. Instead, the project will reduce demand for forest products such firewood collection. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.8 Food			

Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project involves the dissemination of units of cookstoves which does not involve food availability or crop regime or export. Therefore, the safeguarding principle under consideration will not be triggered by the project.	Not Required
>>		The parameter will not be monitored.	
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project involves the dissemination of units of cookstoves which does not involve animal husbandry. Therefore, the safeguarding principle under consideration will not be triggered by the project.	Not Required
>>			

		The parameter will not be monitored.	
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project involves the dissemination of units of cookstoves which does not involve conservation value areas and critical habitats. Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	Not Required
>>			
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	The project involves the dissemination of units of cookstoves which does not involve endangered species.	Not Required

AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		Therefore, the safeguarding principle under consideration will not be triggered by the project. The parameter will not be monitored.	
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Greenway Grameen Infra Private Limited
Registration number with relevant authority	U70102MH2010PTC210549 under Companies Act, 1956, India
Street/P.O. Box	#501-502, Plot No.522, CTS No. F/355, Makani Centre, 35th road, T.P.S. III,
Building	Makani Centre
City	Bandra (W) Mumbai, Mumbai City
State/Region	Maharashtra
Postcode	400050
Country	India
Telephone	+91-22-60522638
E-mail	contact@greenwayappliances.com
Website	https://www.greenwayappliances.com/
Contact person	Mr. Antik Mathur
Title	CEO
Salutation	Mr.
Last name	Mathur
Middle name	-
First name	Antik
Department	Management
Mobile	+91 7303319269
Direct tel.	+91 7303319269
Personal e-mail	ankit@greenwayappliances.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of [Principles and Requirements](#) for more information on procedures governing Design Changes

None.

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption